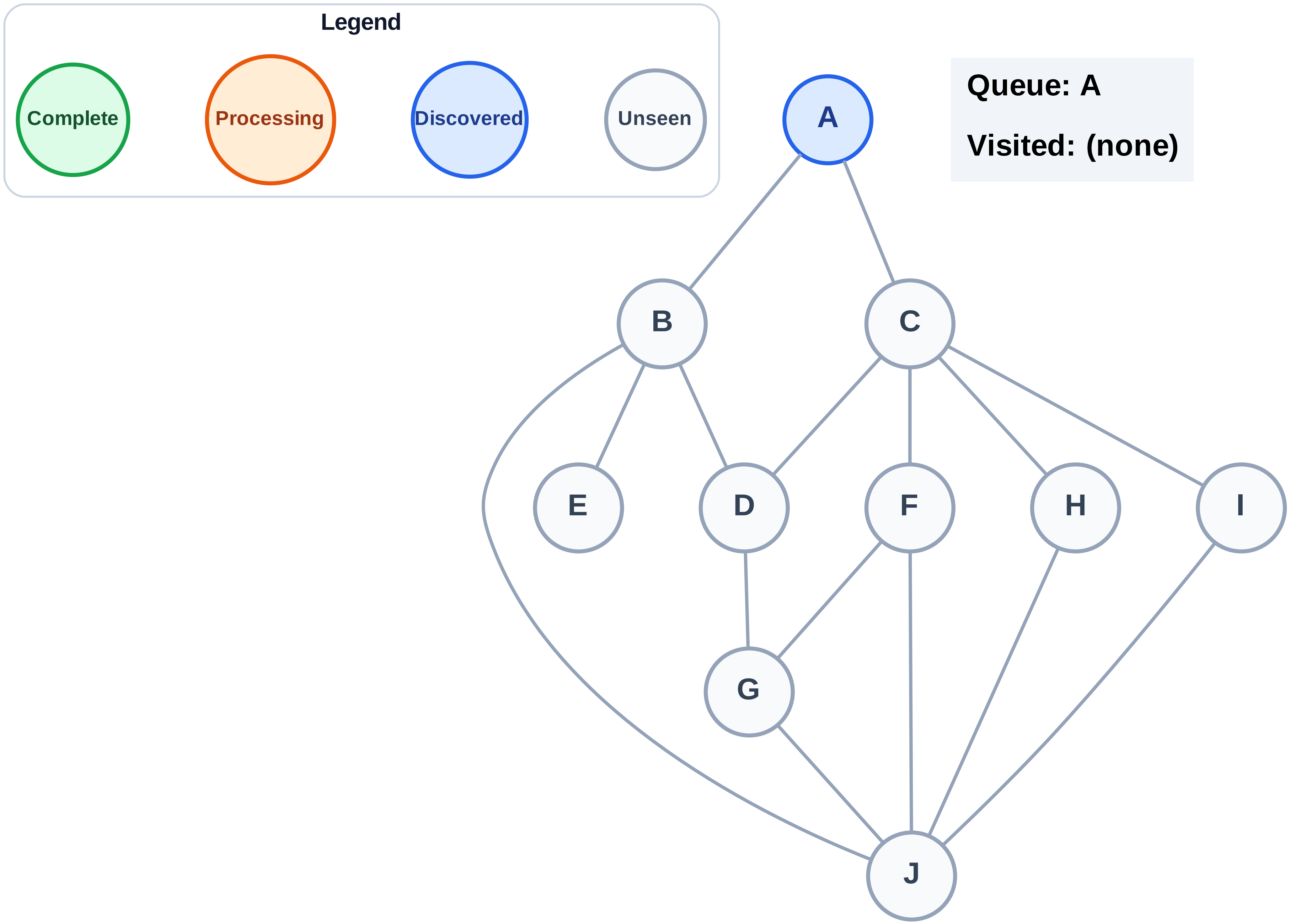
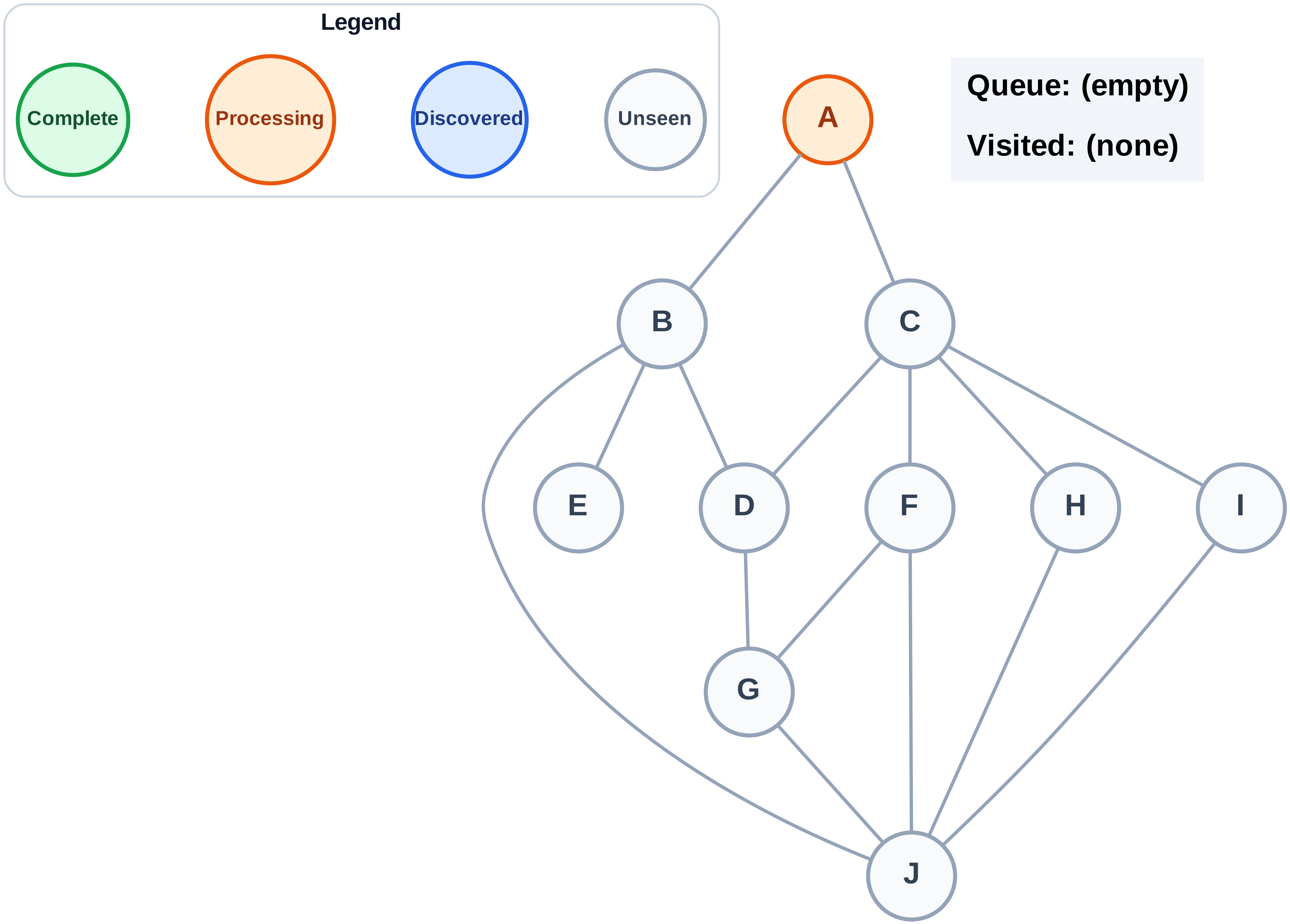
BFS Traversal

Step 1: Start at A



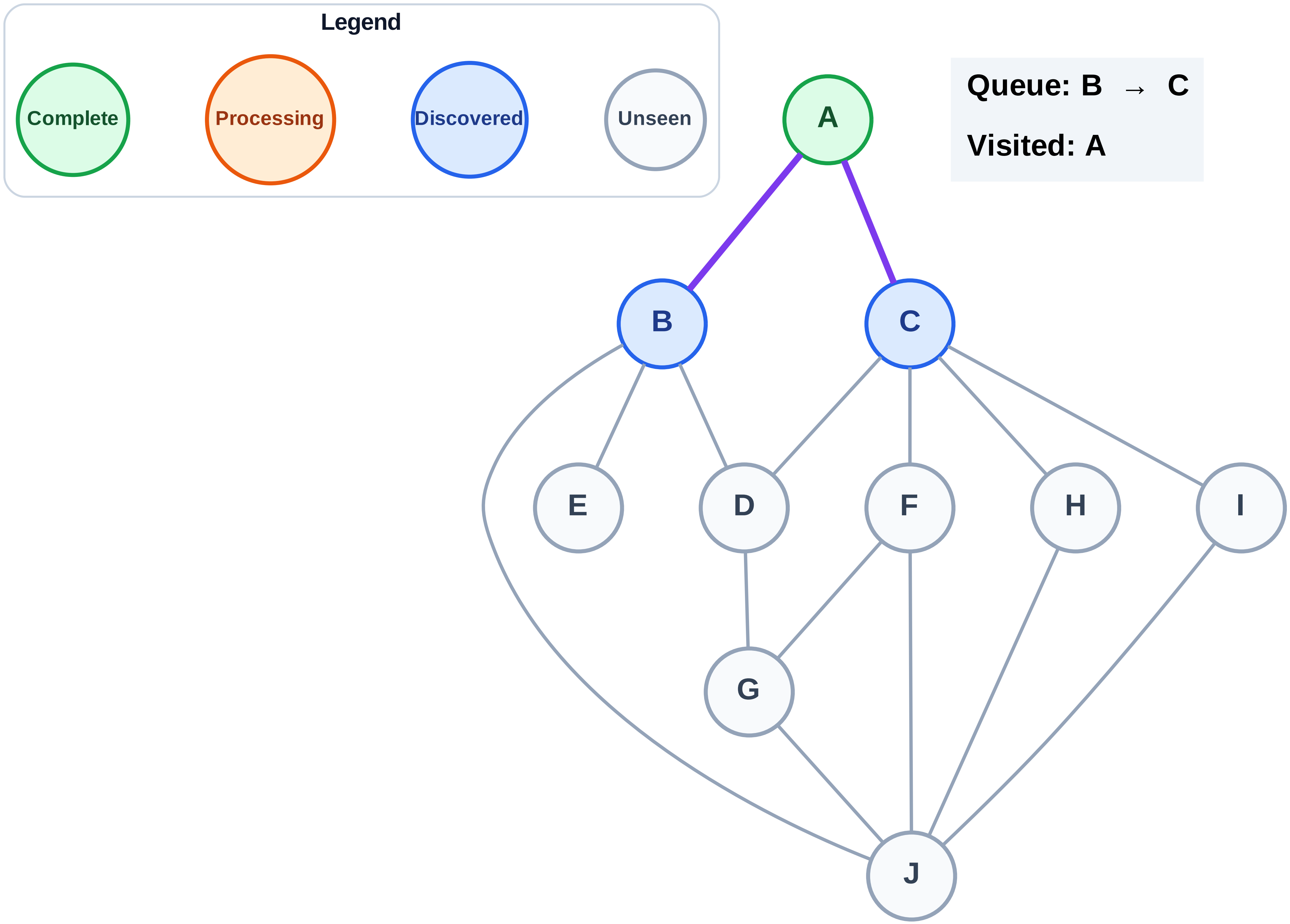
BFS Traversal

Step 2: Process node A



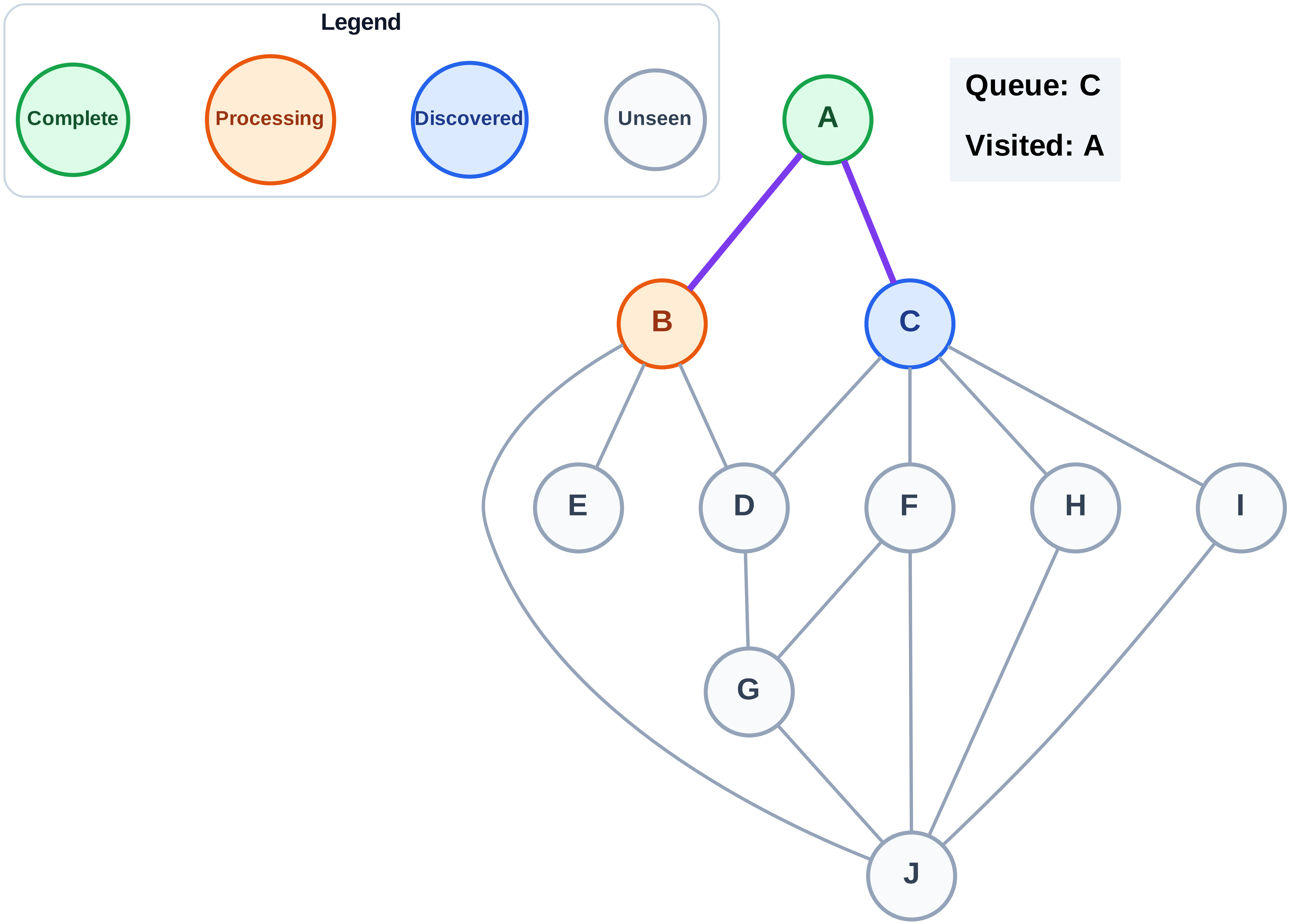
BFS Traversal

Step 3: Discover B, C from A



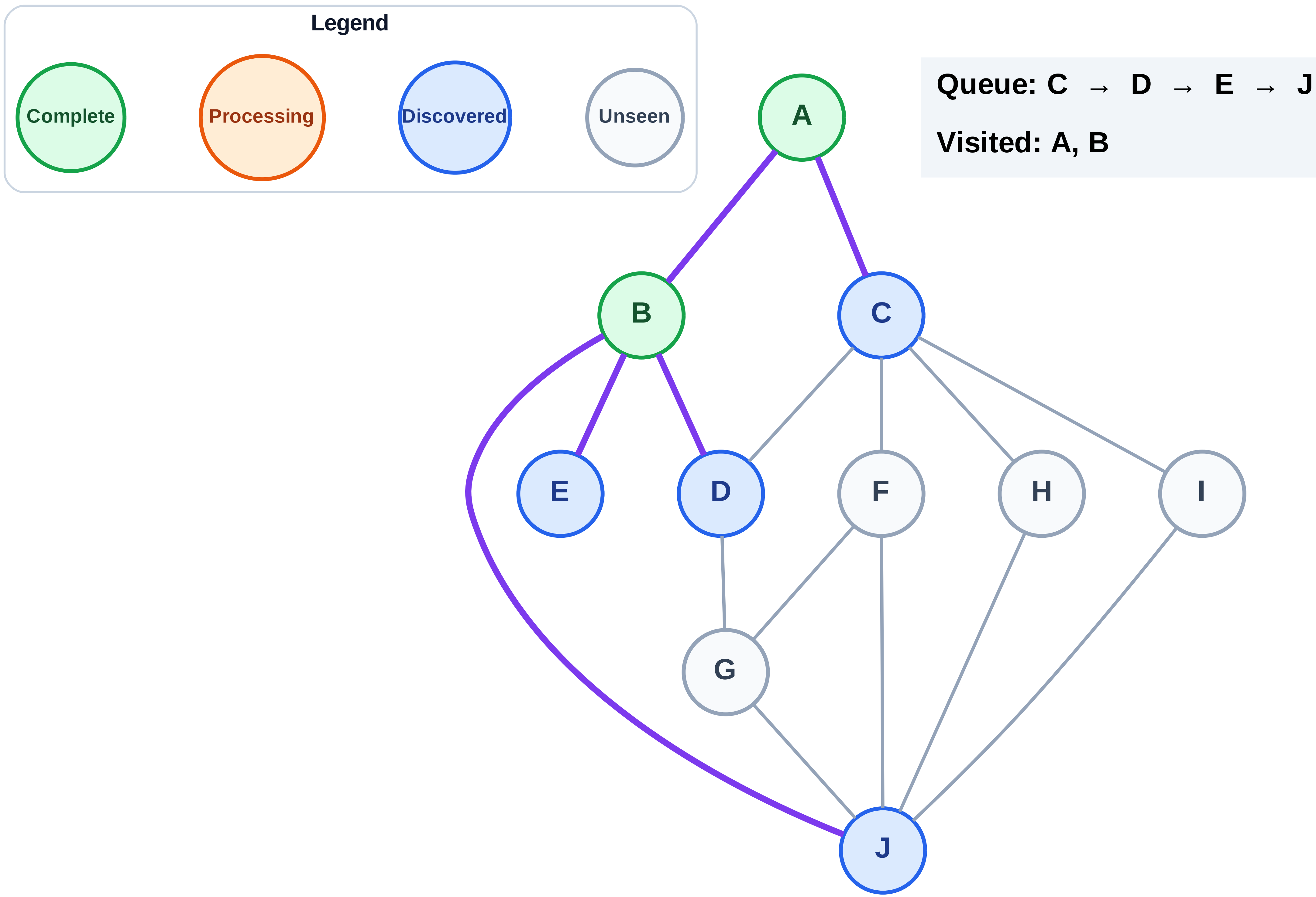
BFS Traversal

Step 4: Process node B



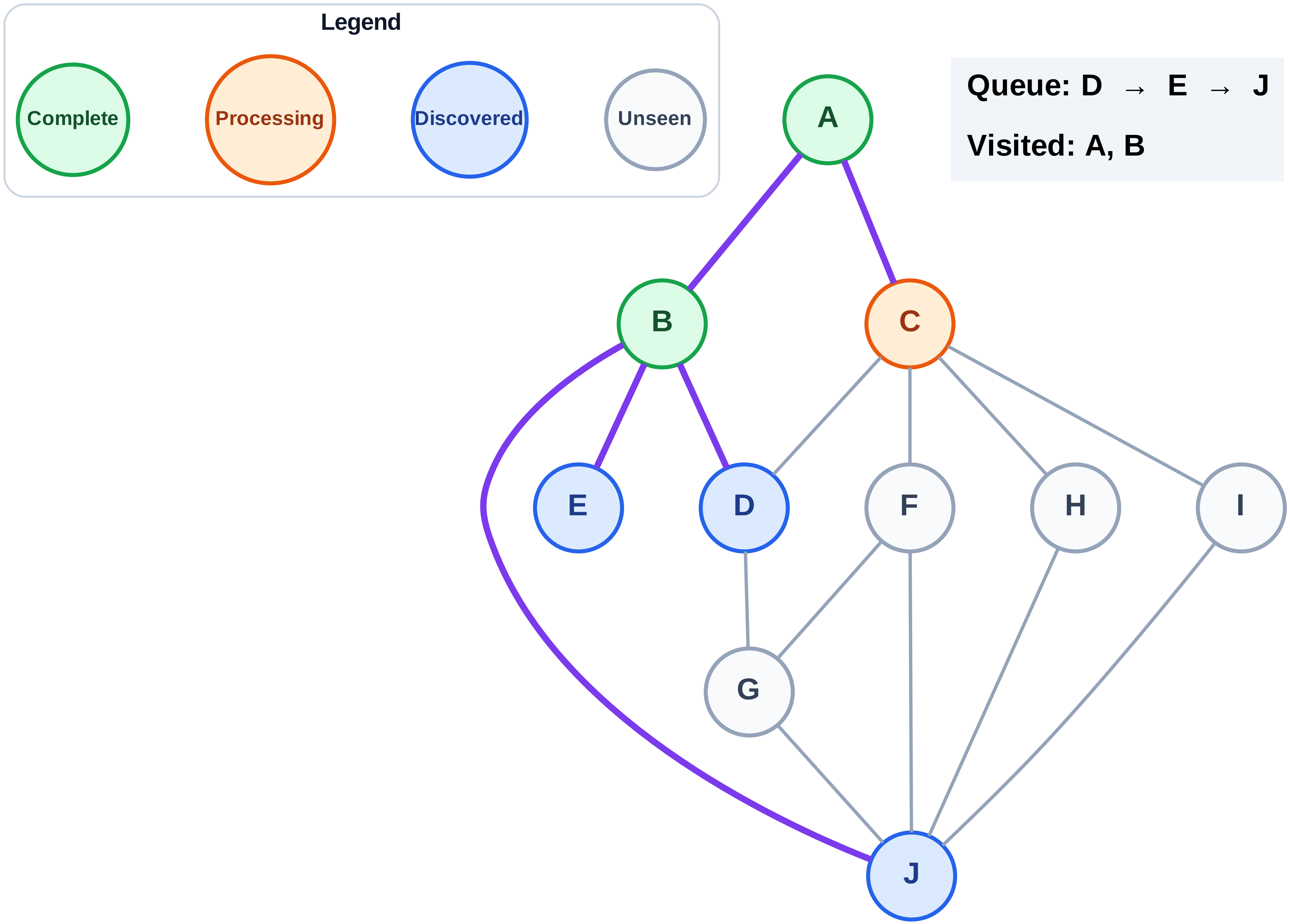
BFS Traversal

Step 5: Discover D, E, J from B



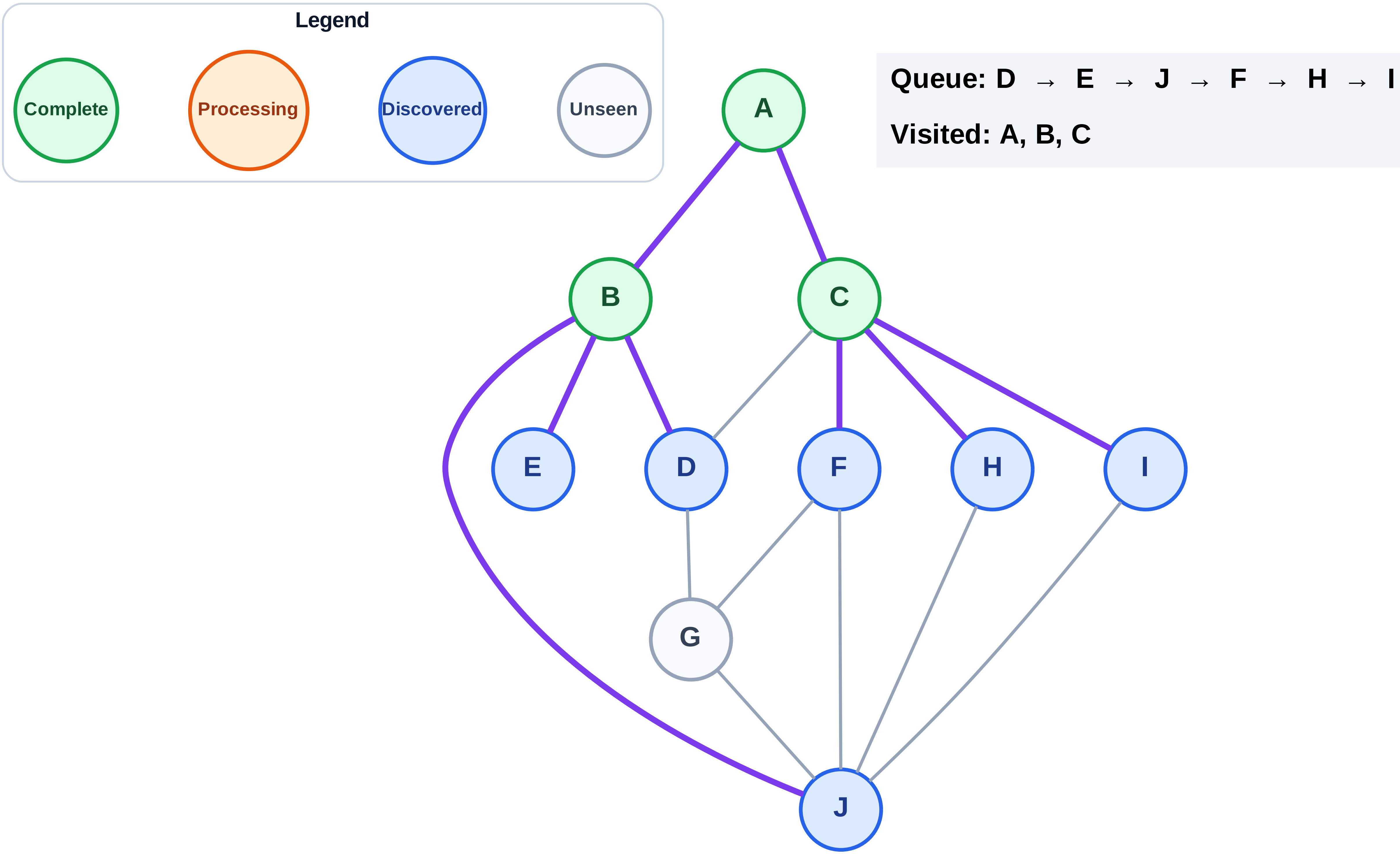
BFS Traversal

Step 6: Process node C



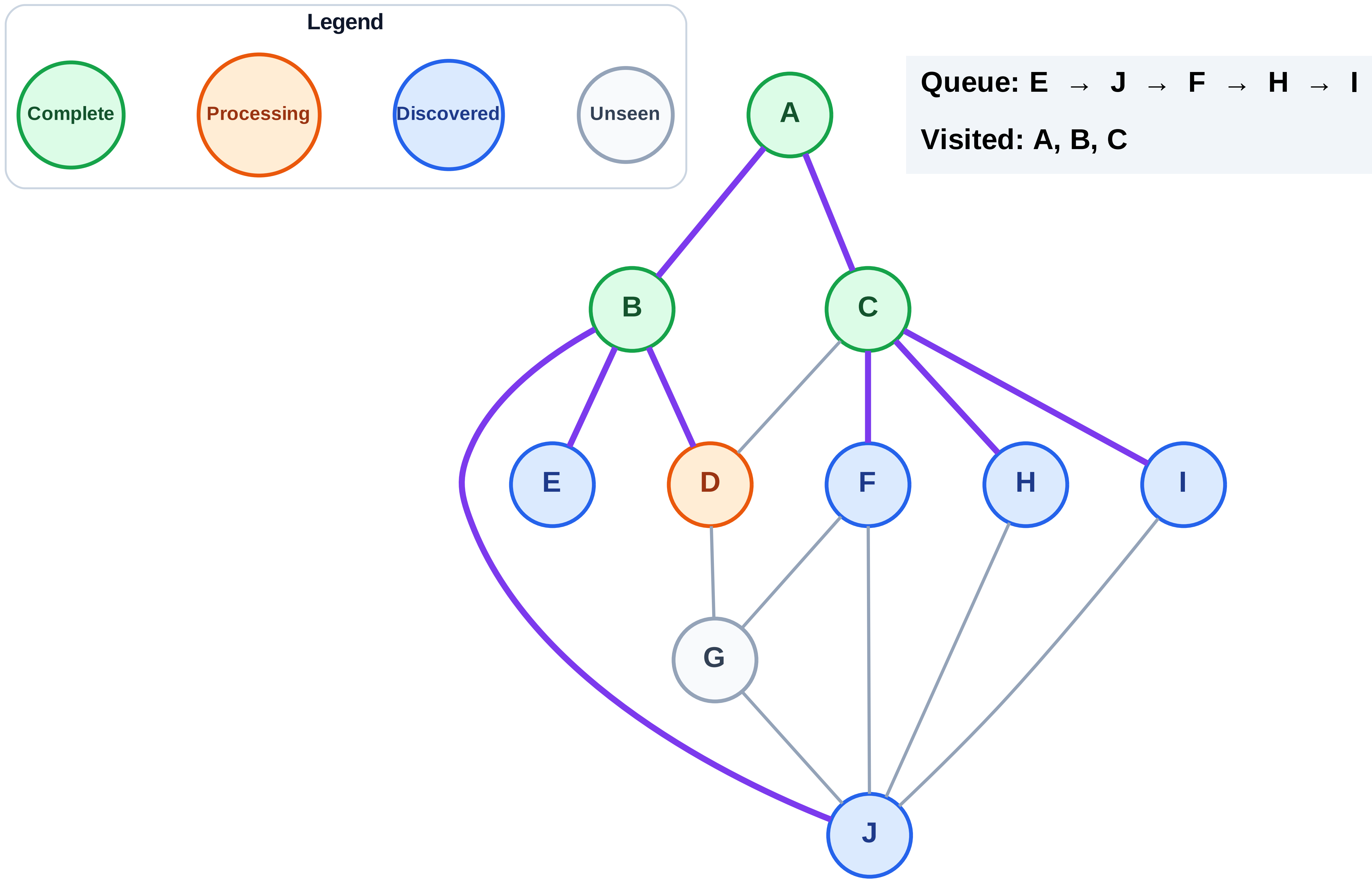
BFS Traversal

Step 7: Discover F, H, I from C



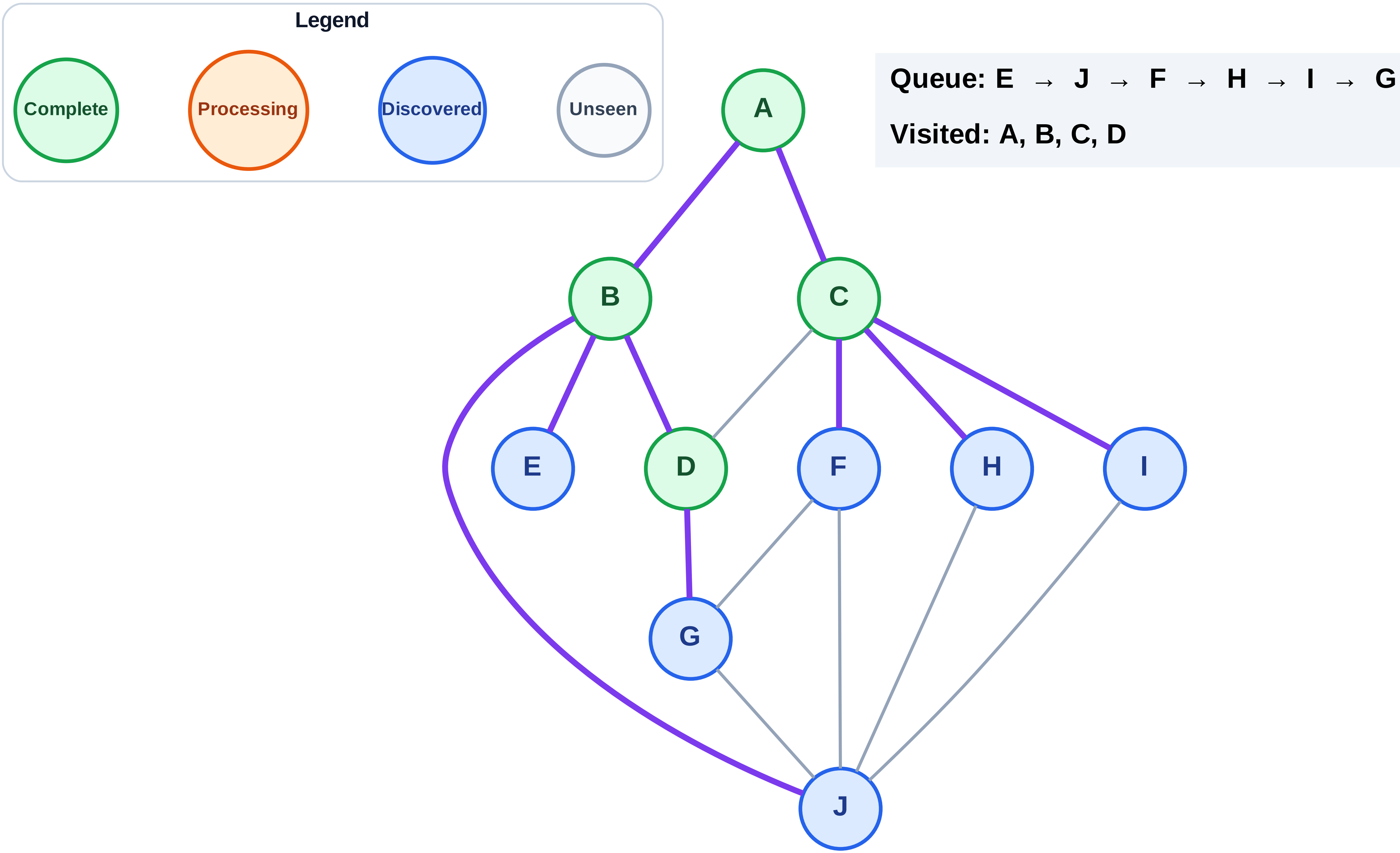
BFS Traversal

Step 8: Process node D



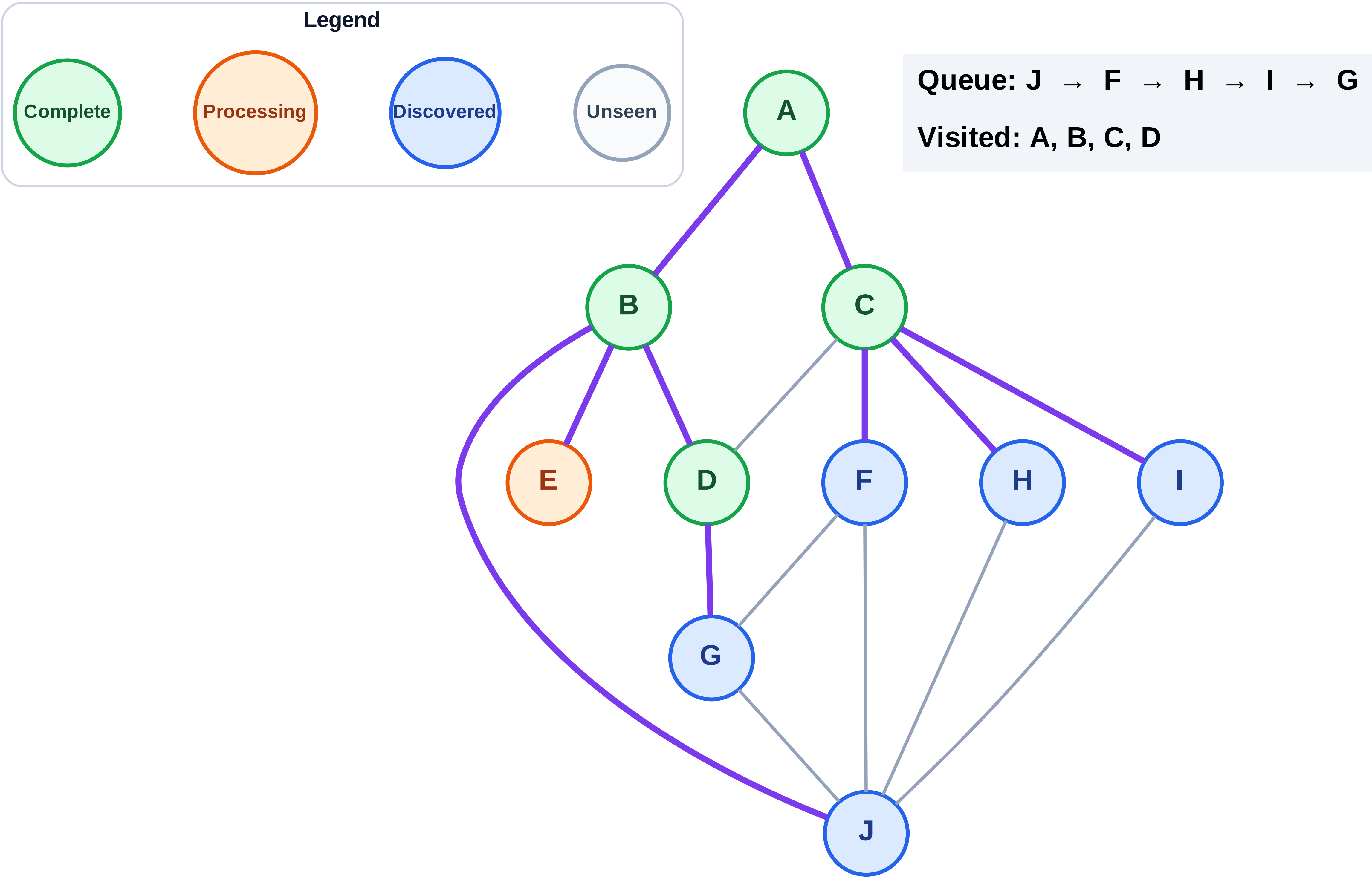
BFS Traversal

Step 9: Discover G from D



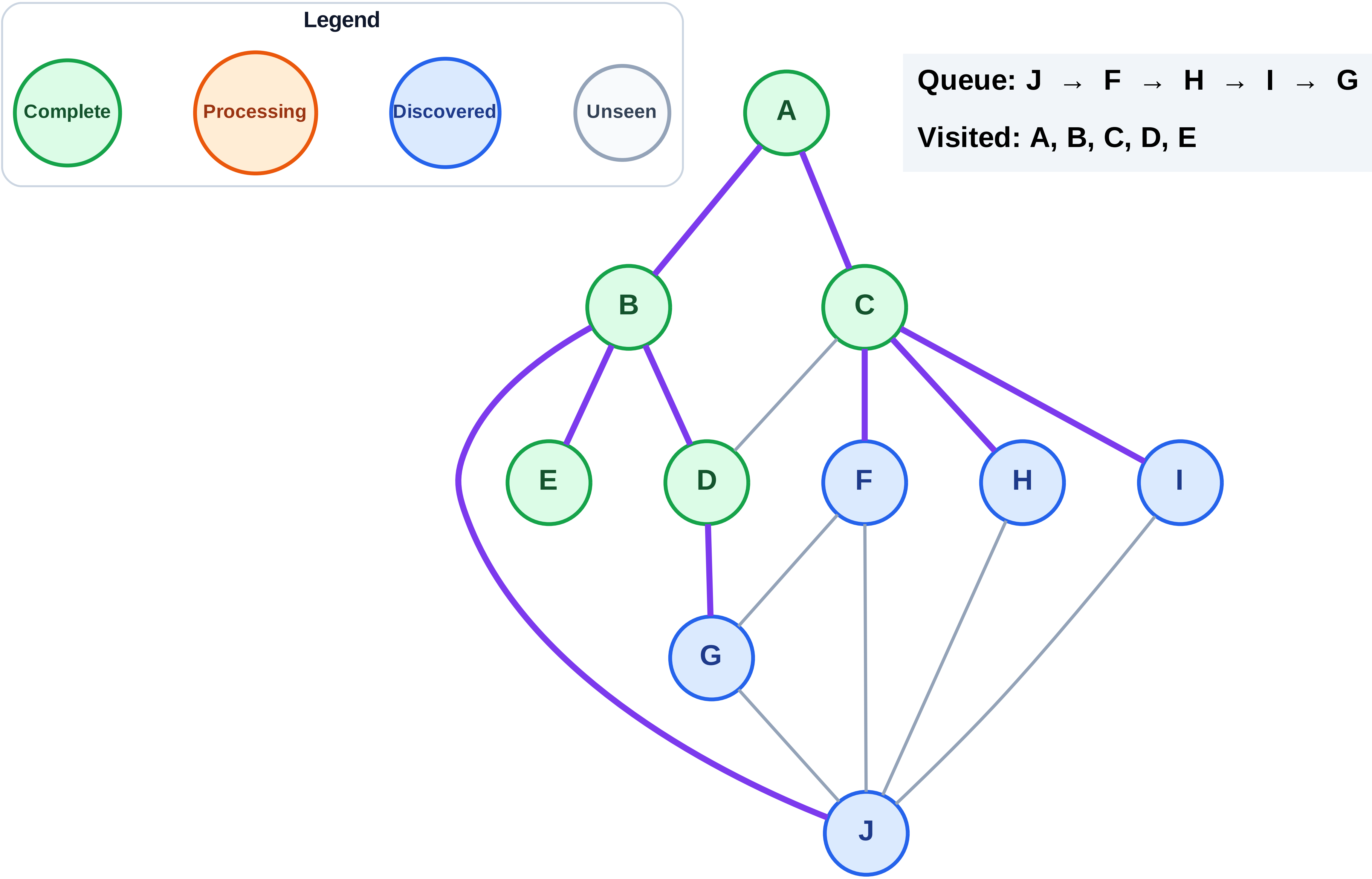
BFS Traversal

Step 10: Process node E



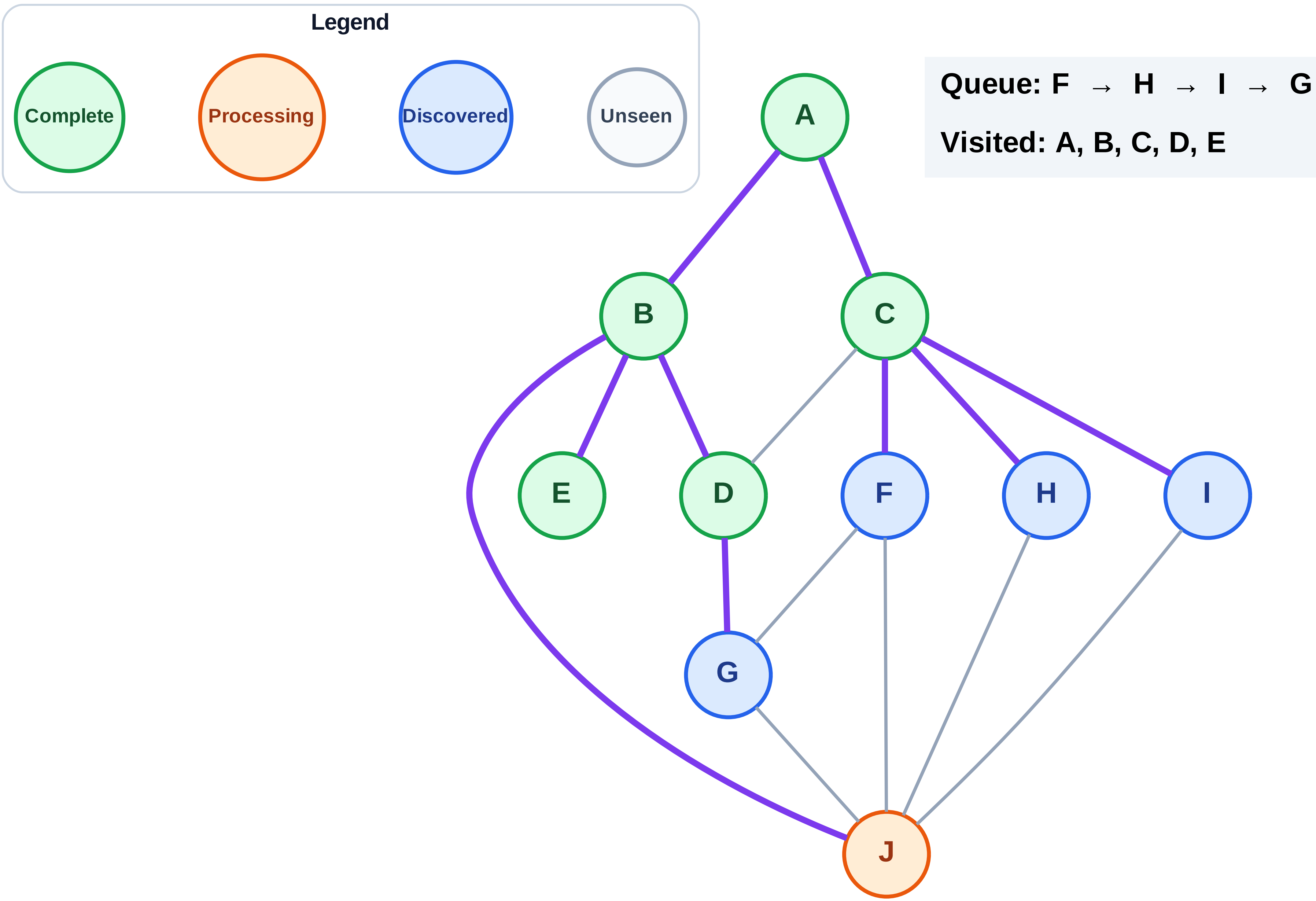
BFS Traversal

Step 11: No new neighbors from E



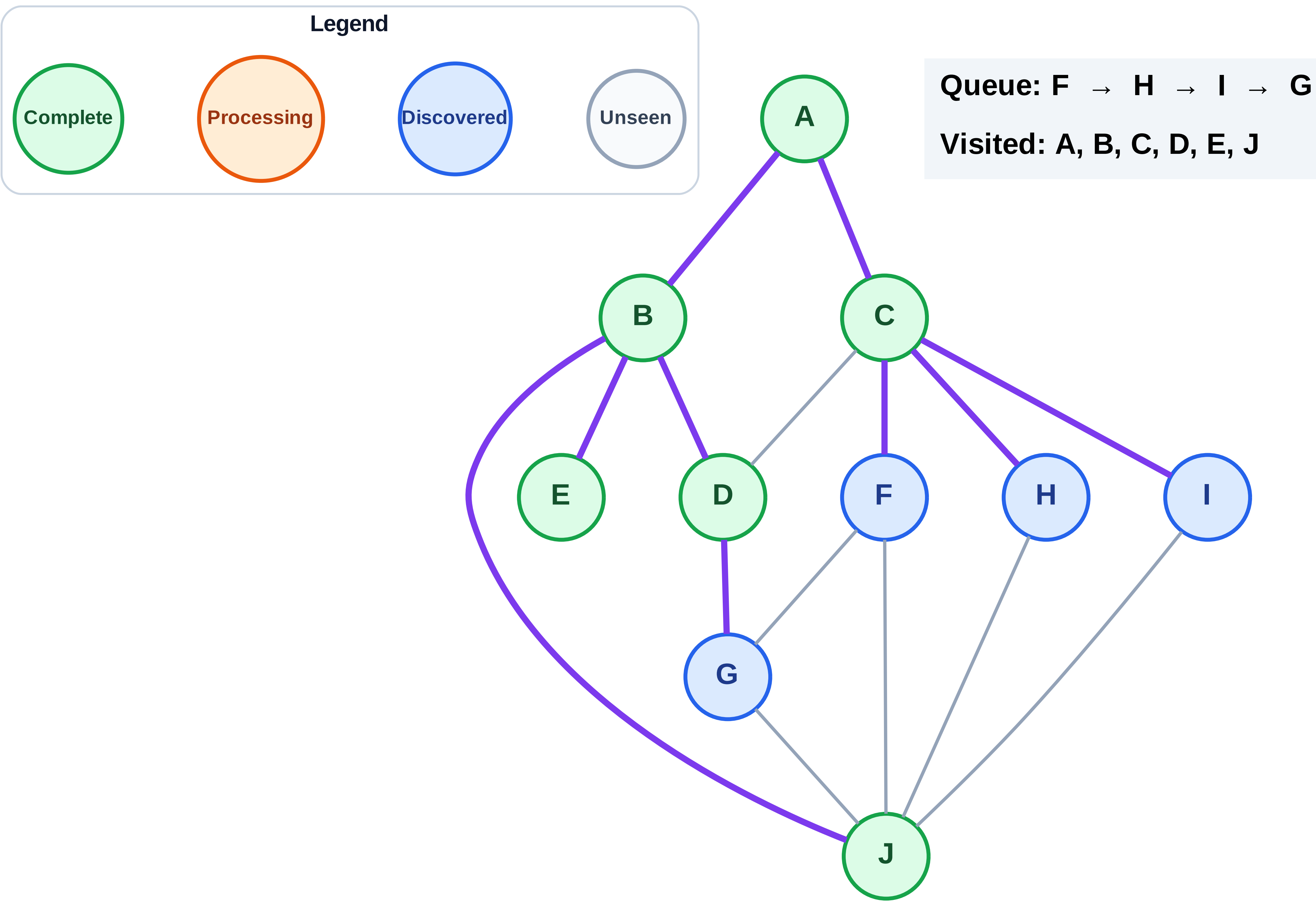
BFS Traversal

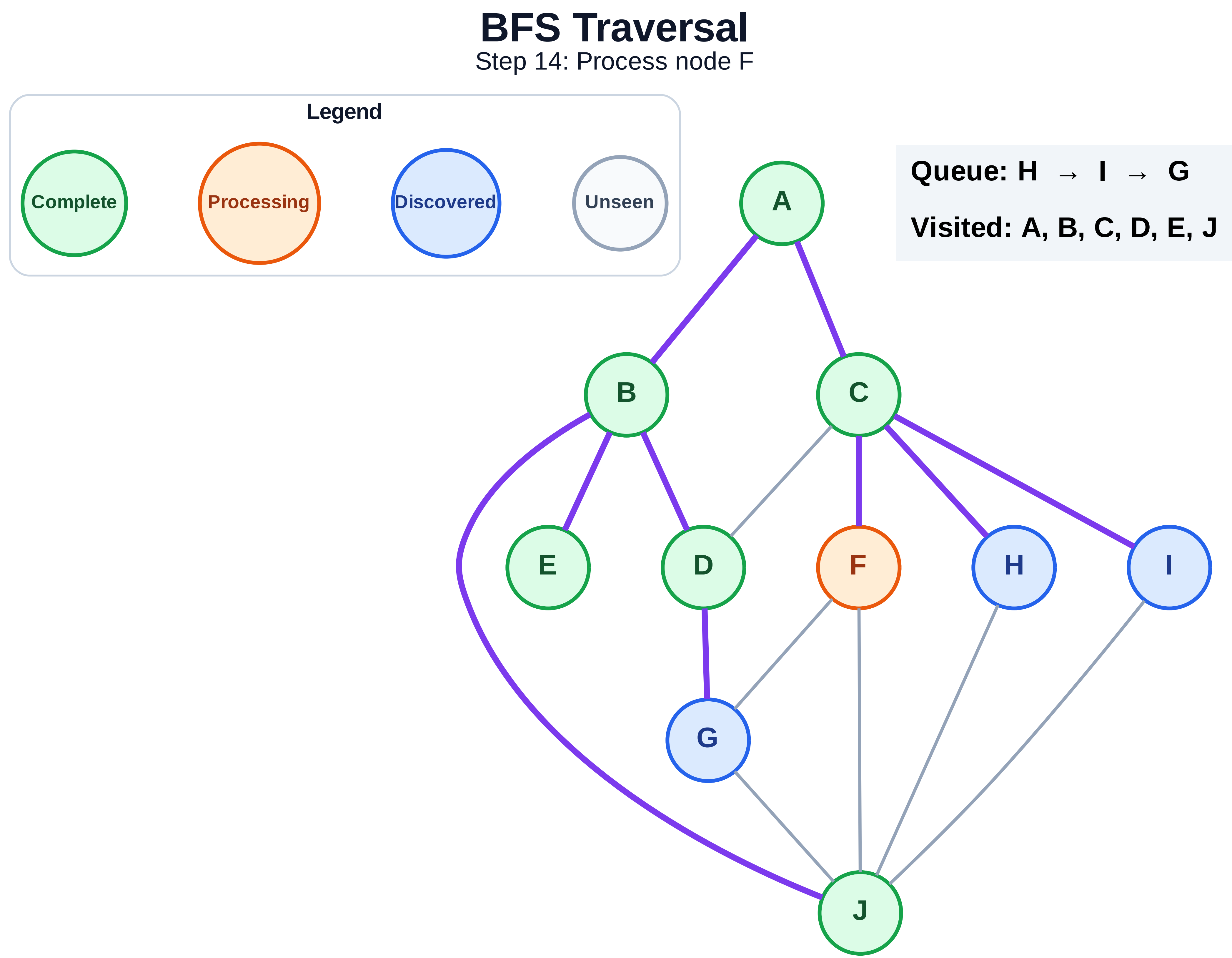
Step 12: Process node J

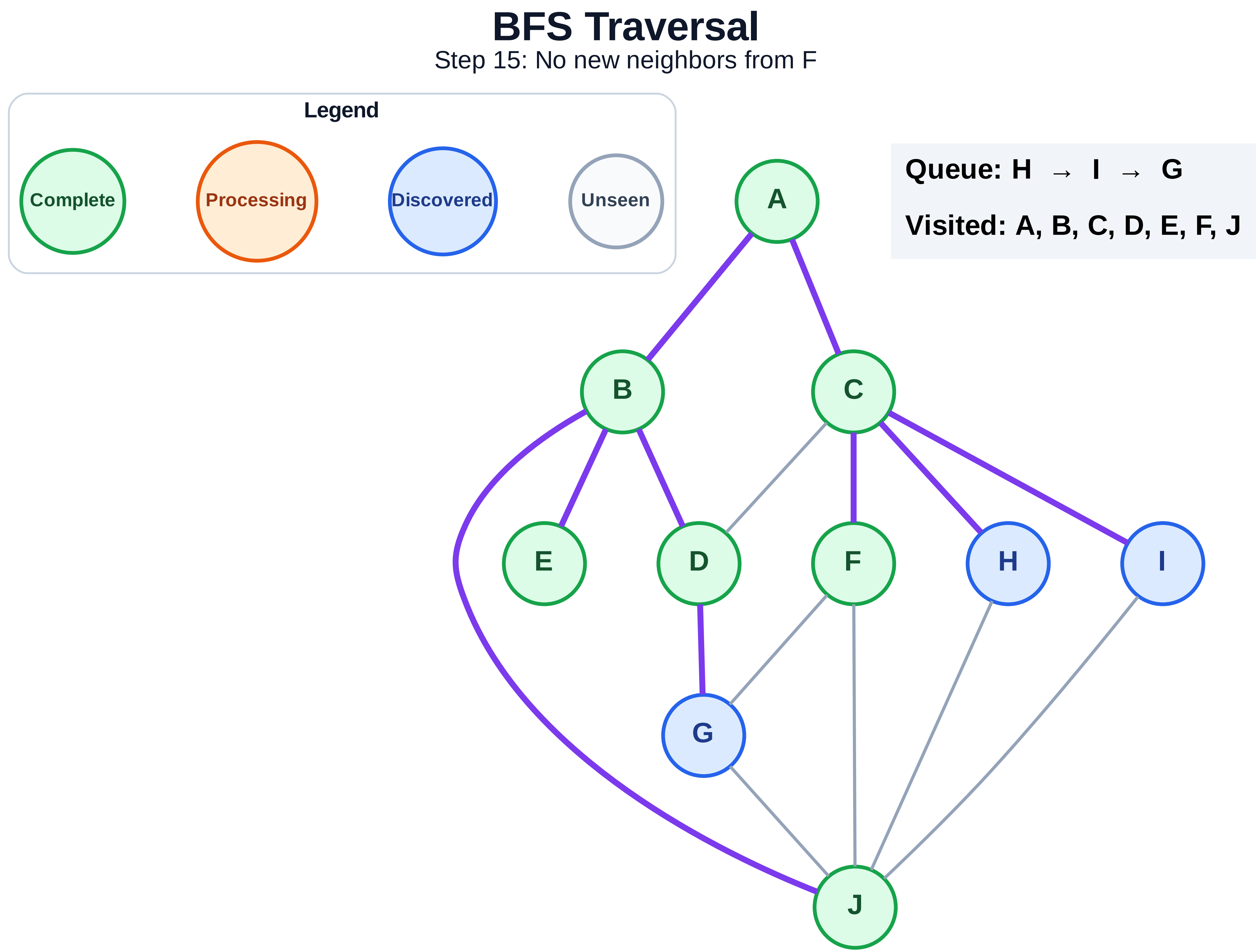


BFS Traversal

Step 13: No new neighbors from J



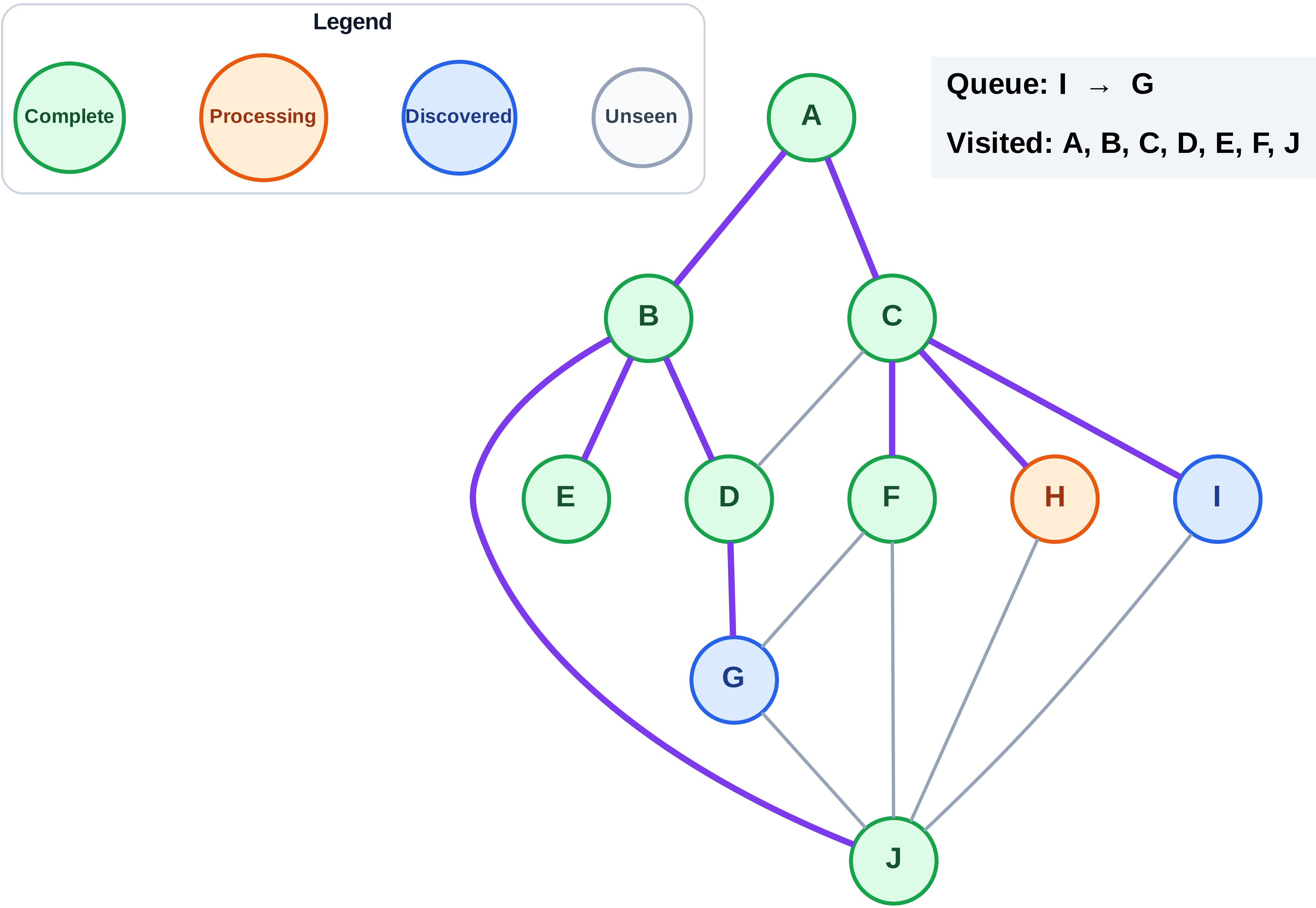

BFS Traversal
Step 14: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: H → I → G
Visited: A, B, C, D, E, J
A
B
C
E
D
F
H
I
G
J



BFS Traversal
Step 15: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Queue: H → I → G
Visited: A, B, C, D, E, F, J
A
B
C
E
D
F
H
I
G
J

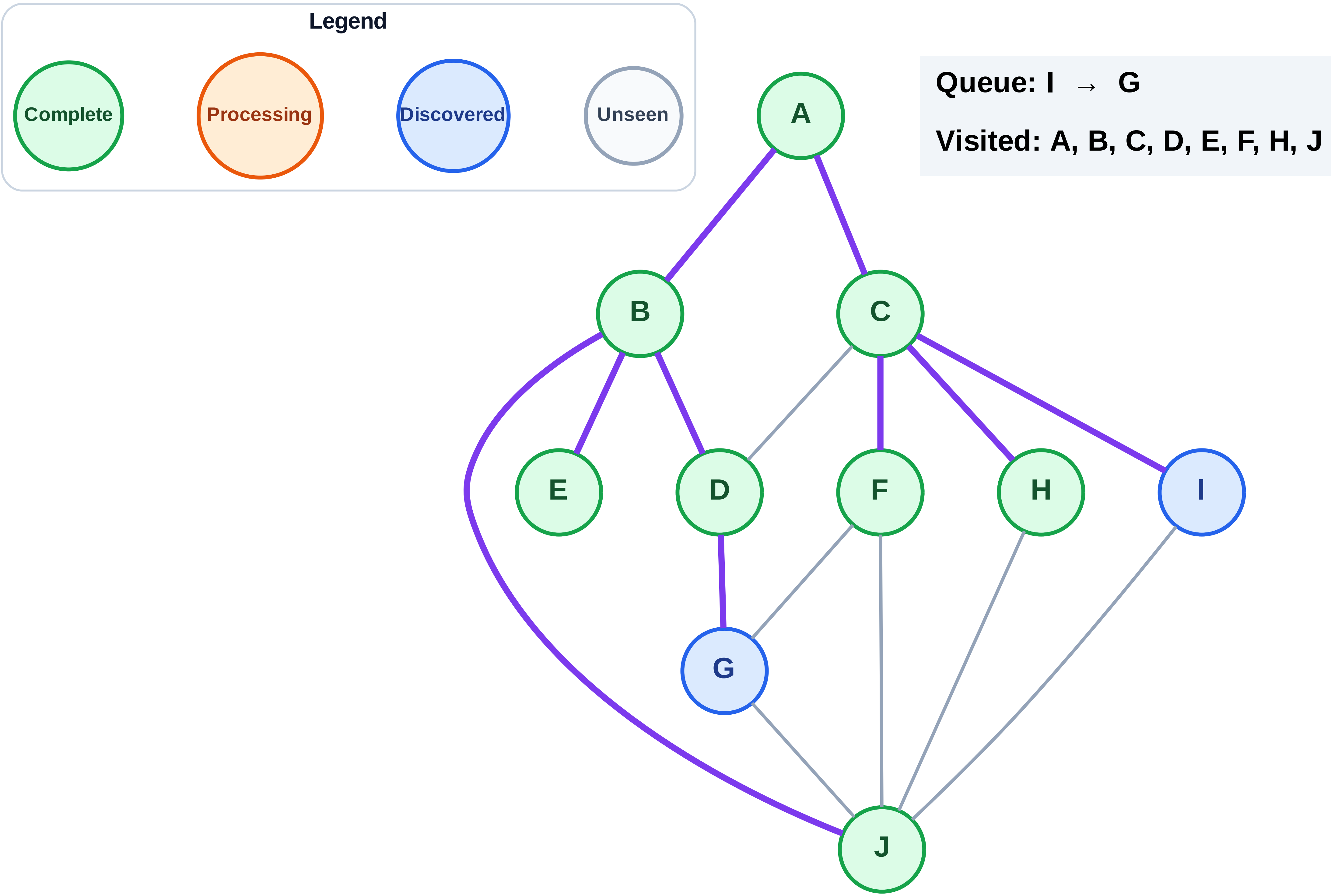
BFS Traversal

Step 16: Process node H



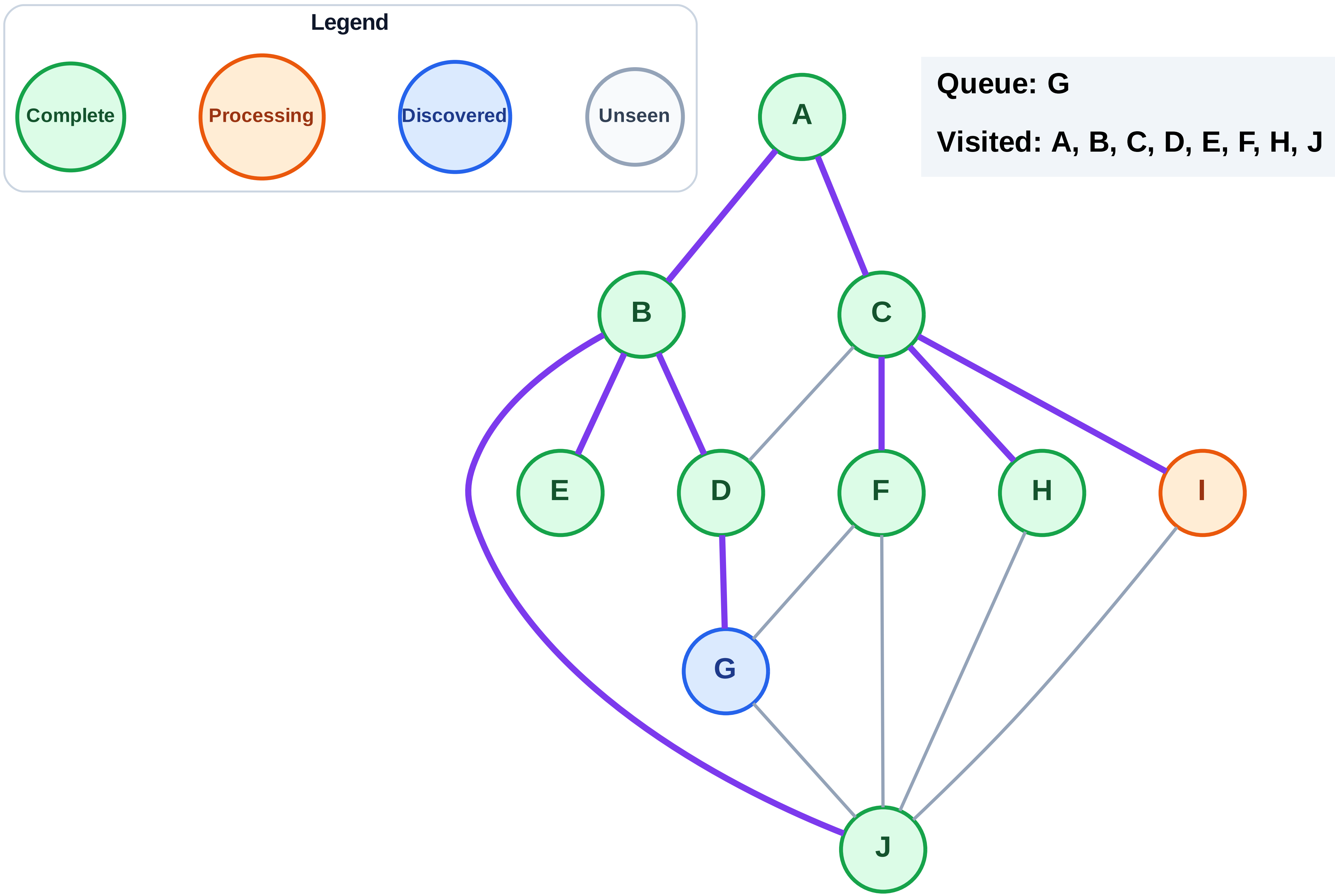
BFS Traversal

Step 17: No new neighbors from H



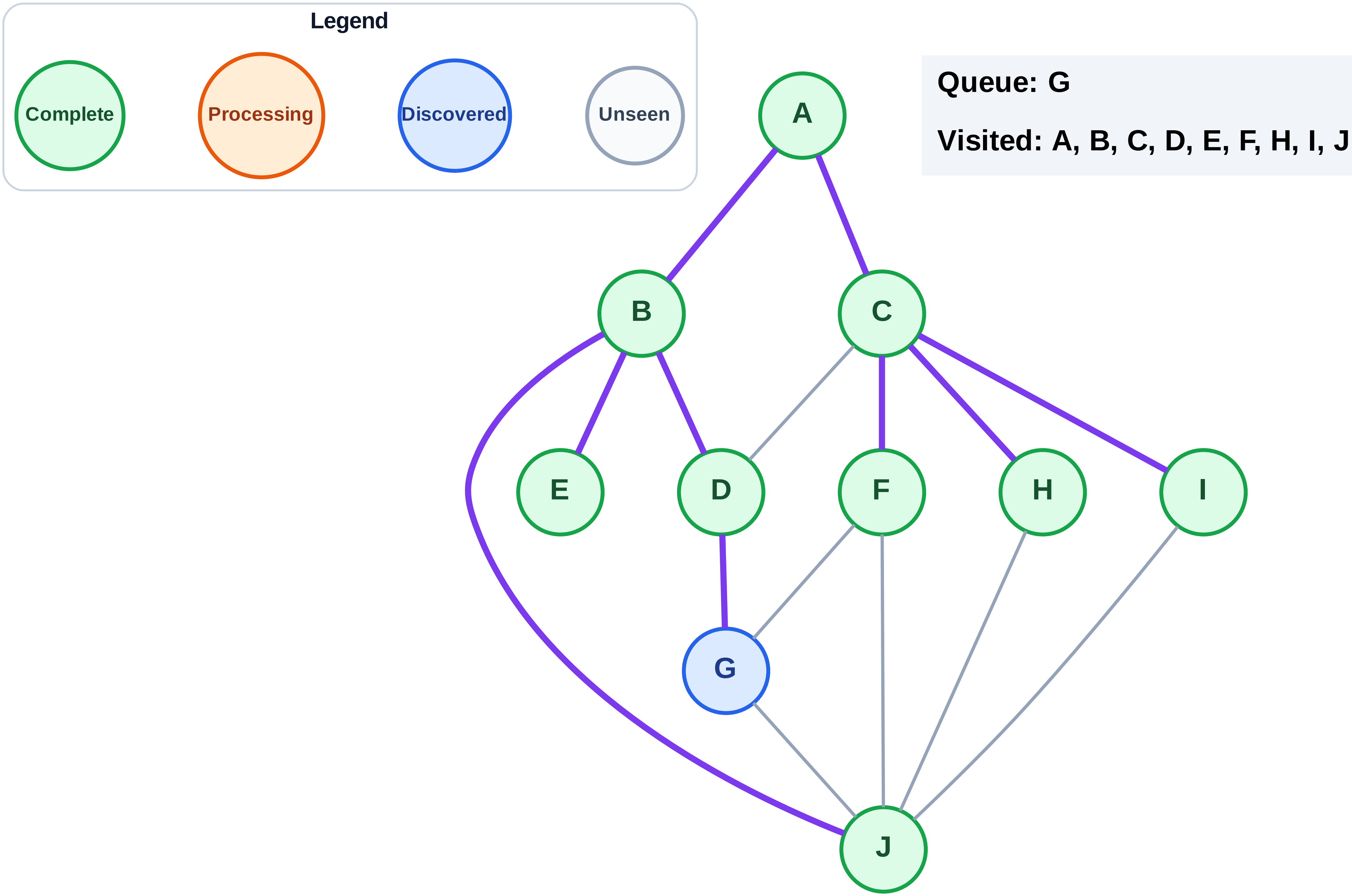
BFS Traversal

Step 18: Process node I



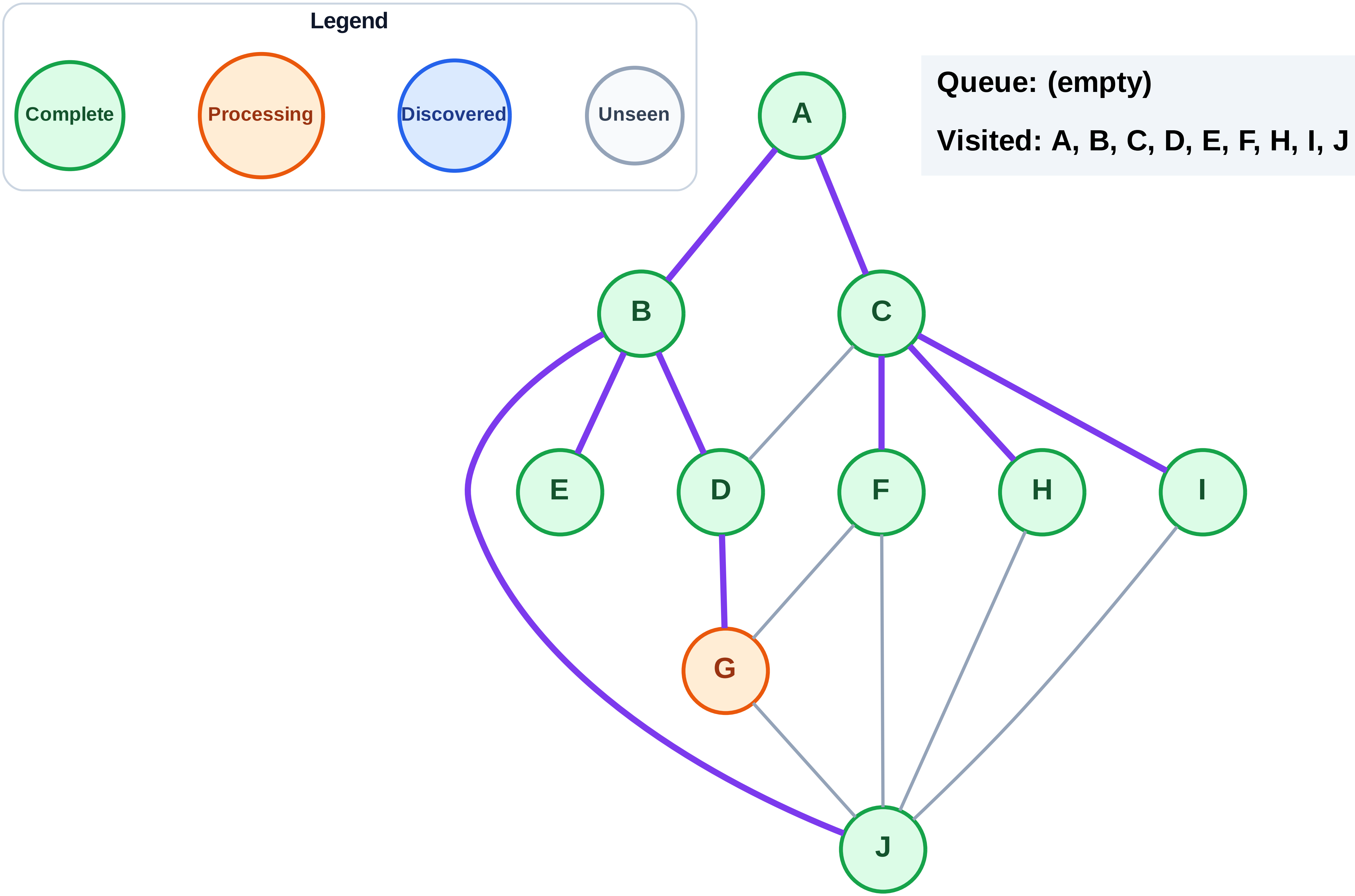
BFS Traversal

Step 19: No new neighbors from I



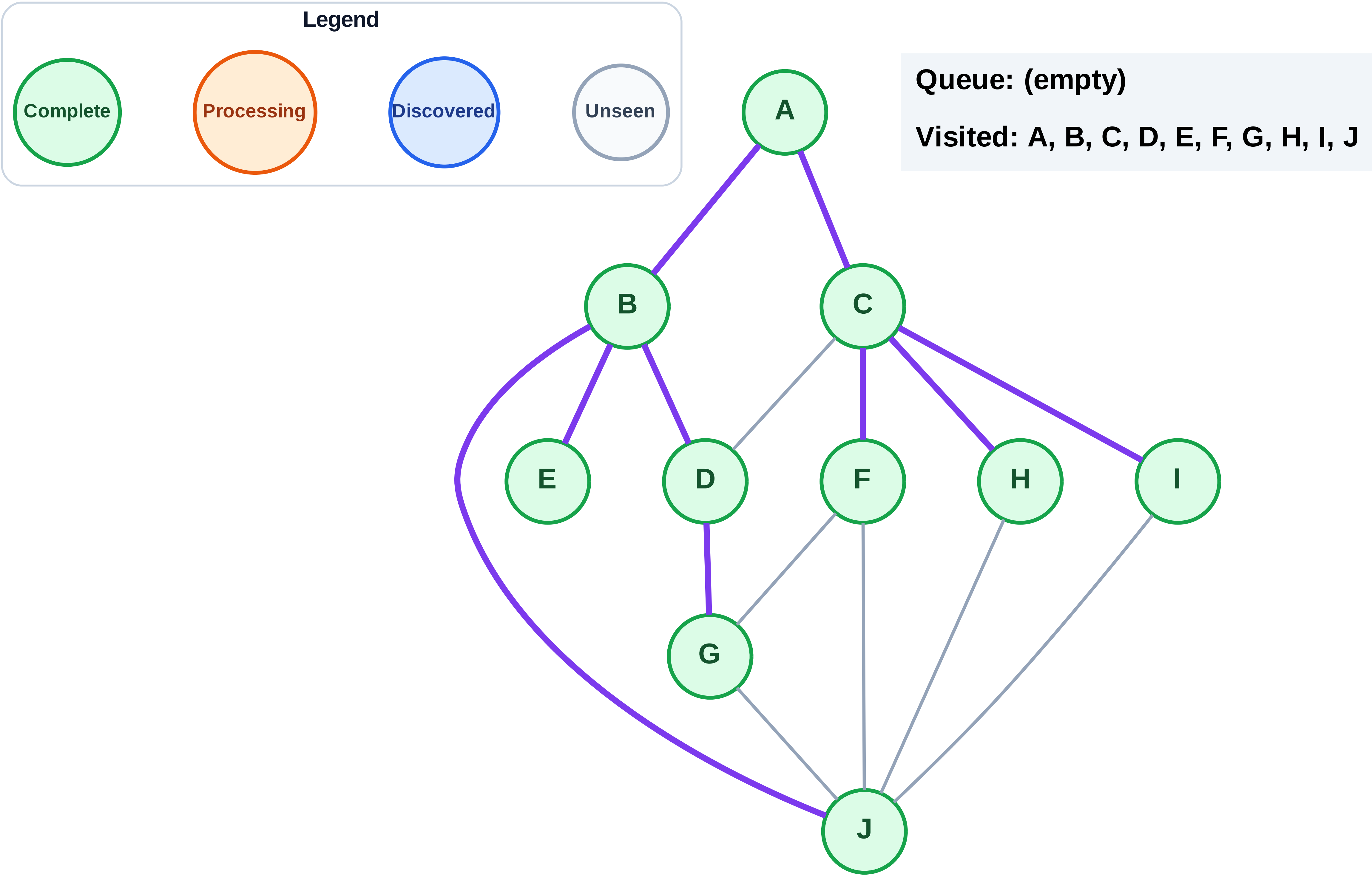
BFS Traversal

Step 20: Process node G



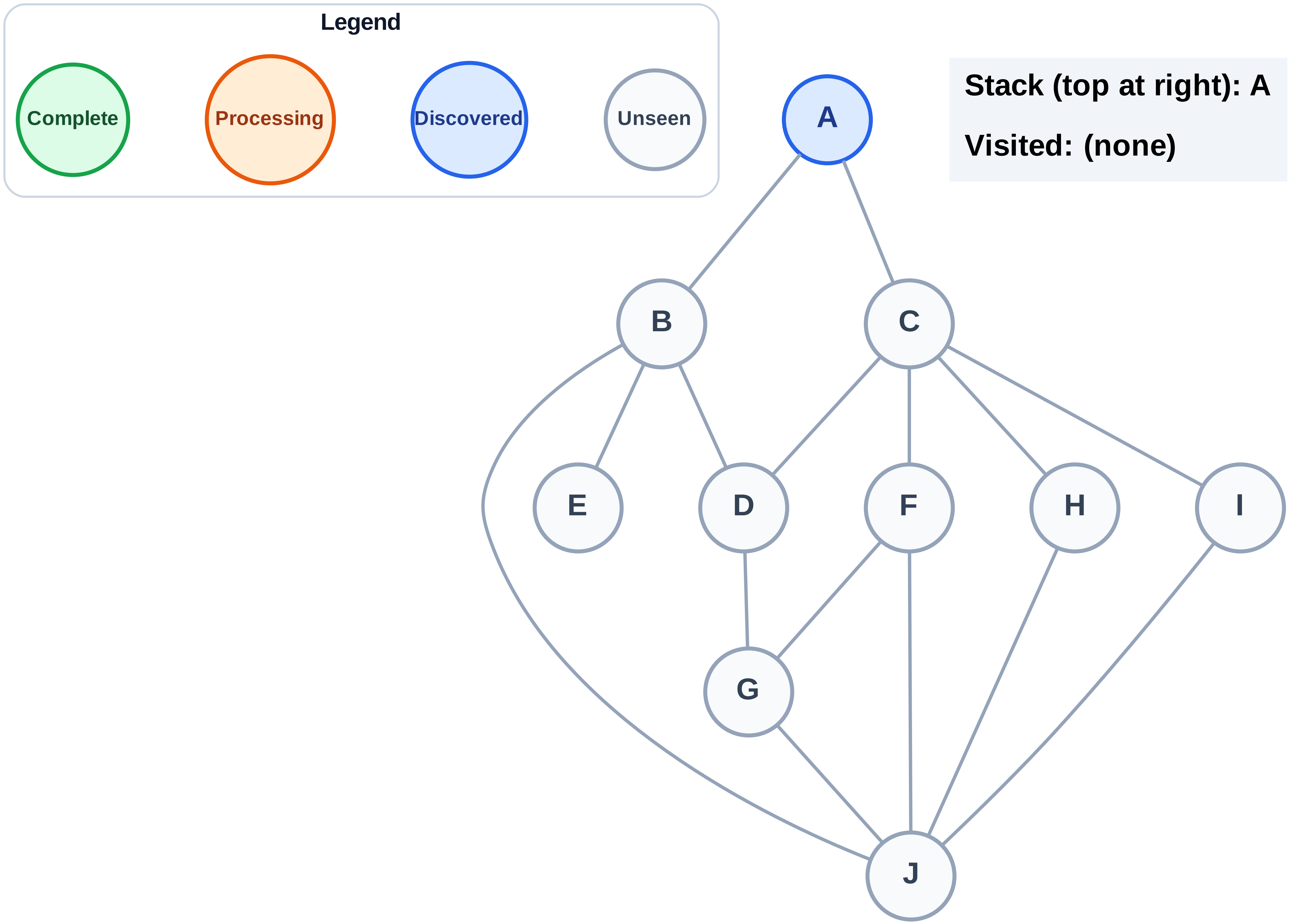
BFS Traversal

Step 21: No new neighbors from G



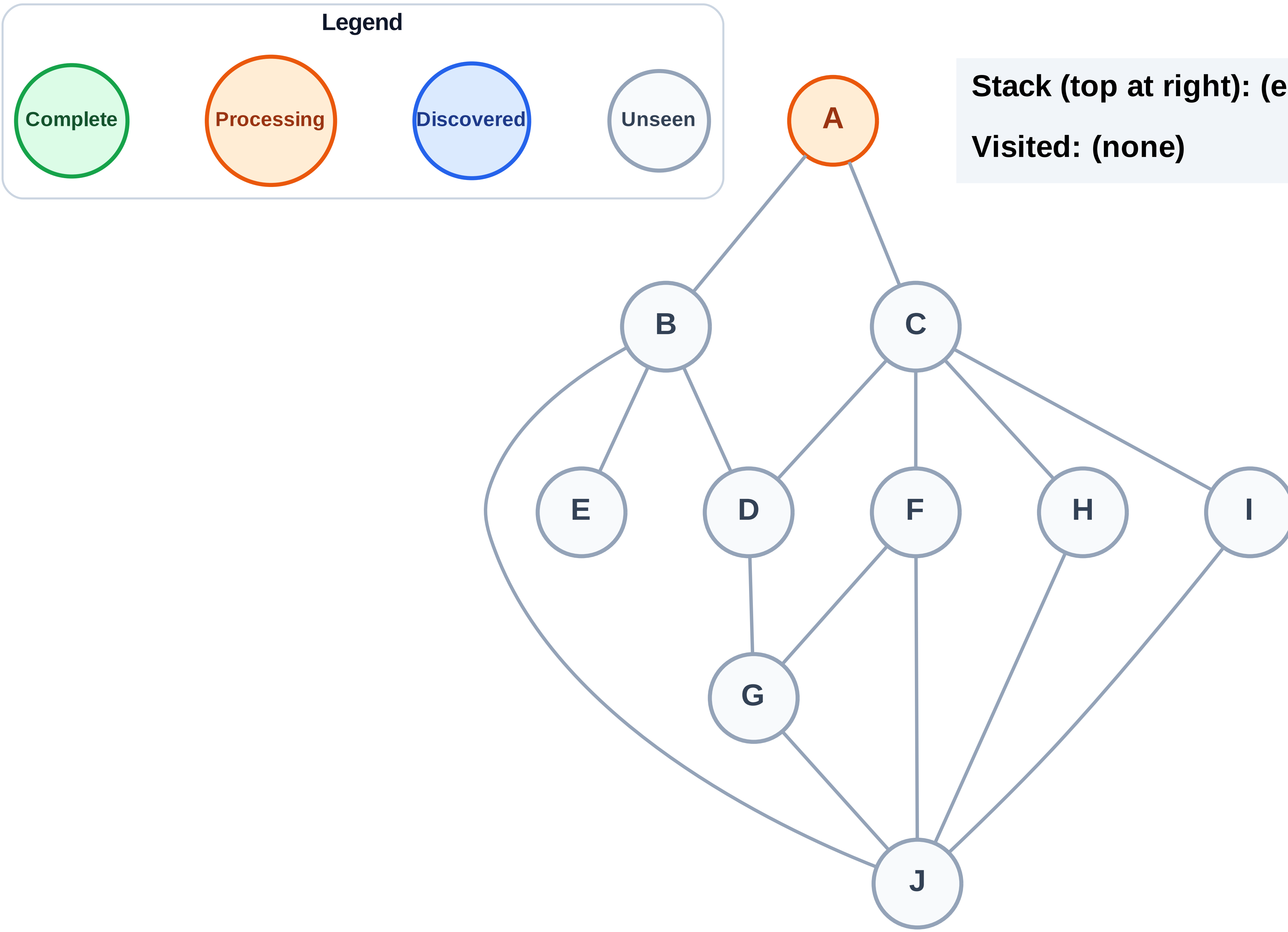
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

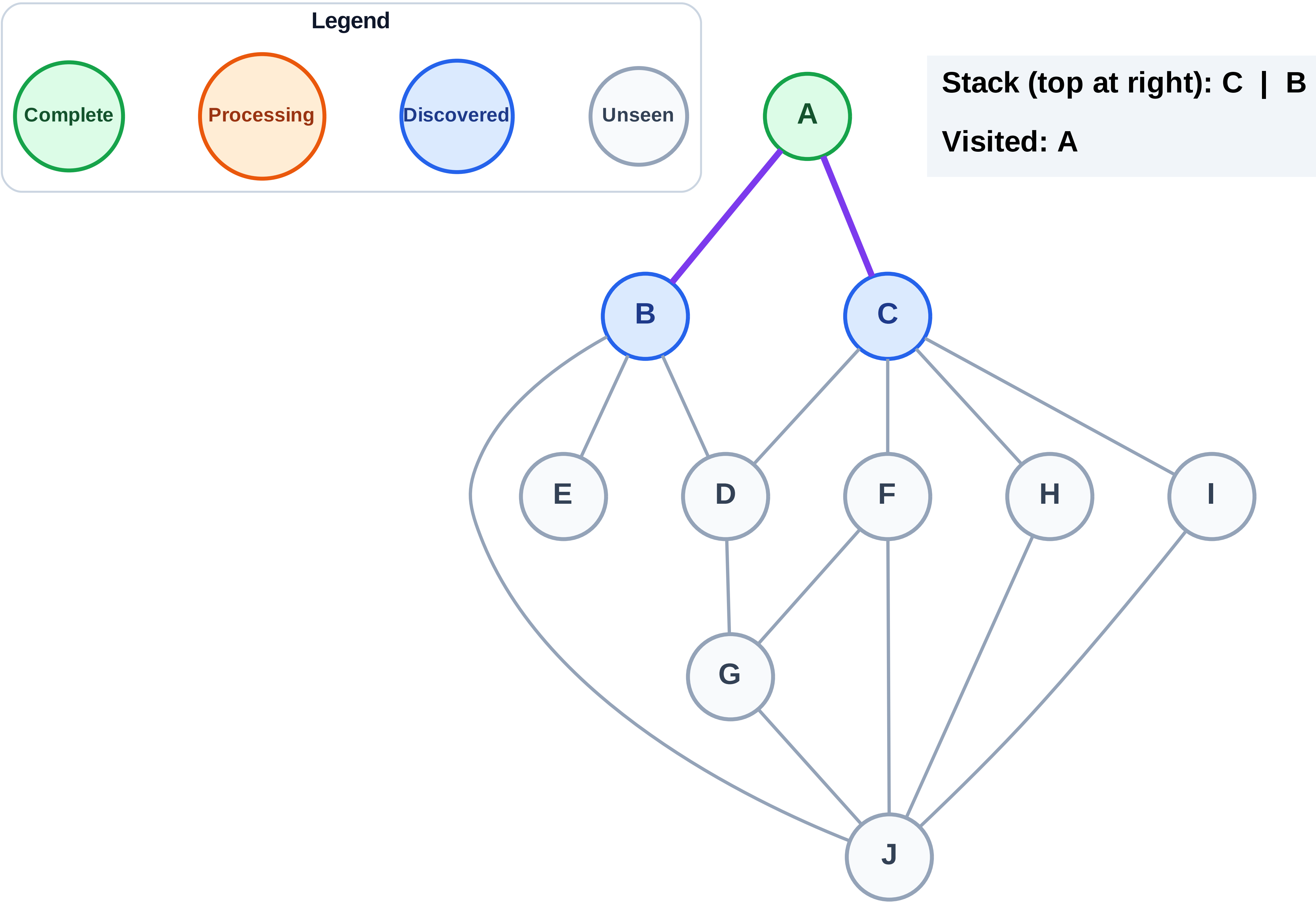


Stack (top at right): (empty)

Visited: (none)

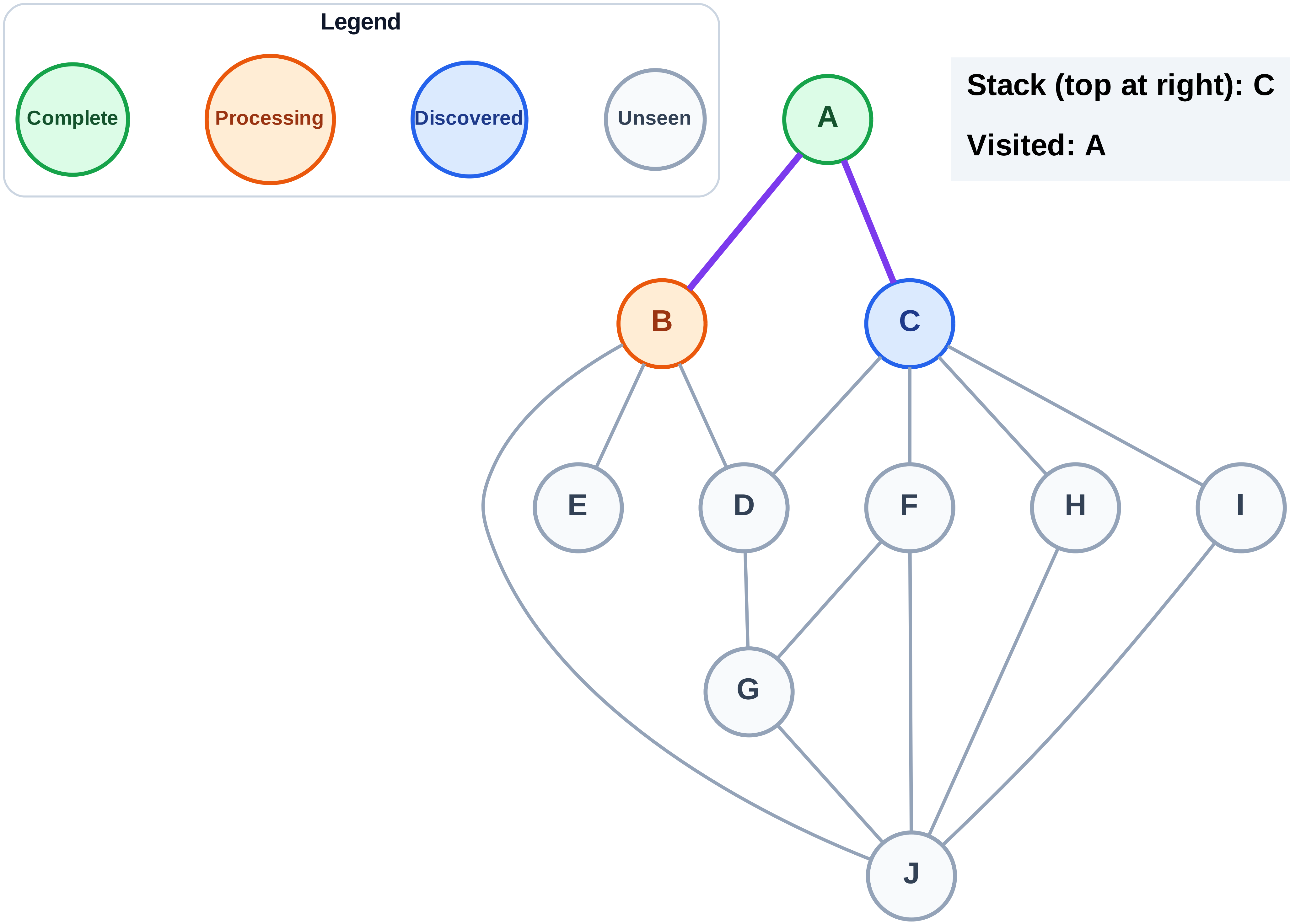
DFS Traversal

Step 3: Discover C, B from A



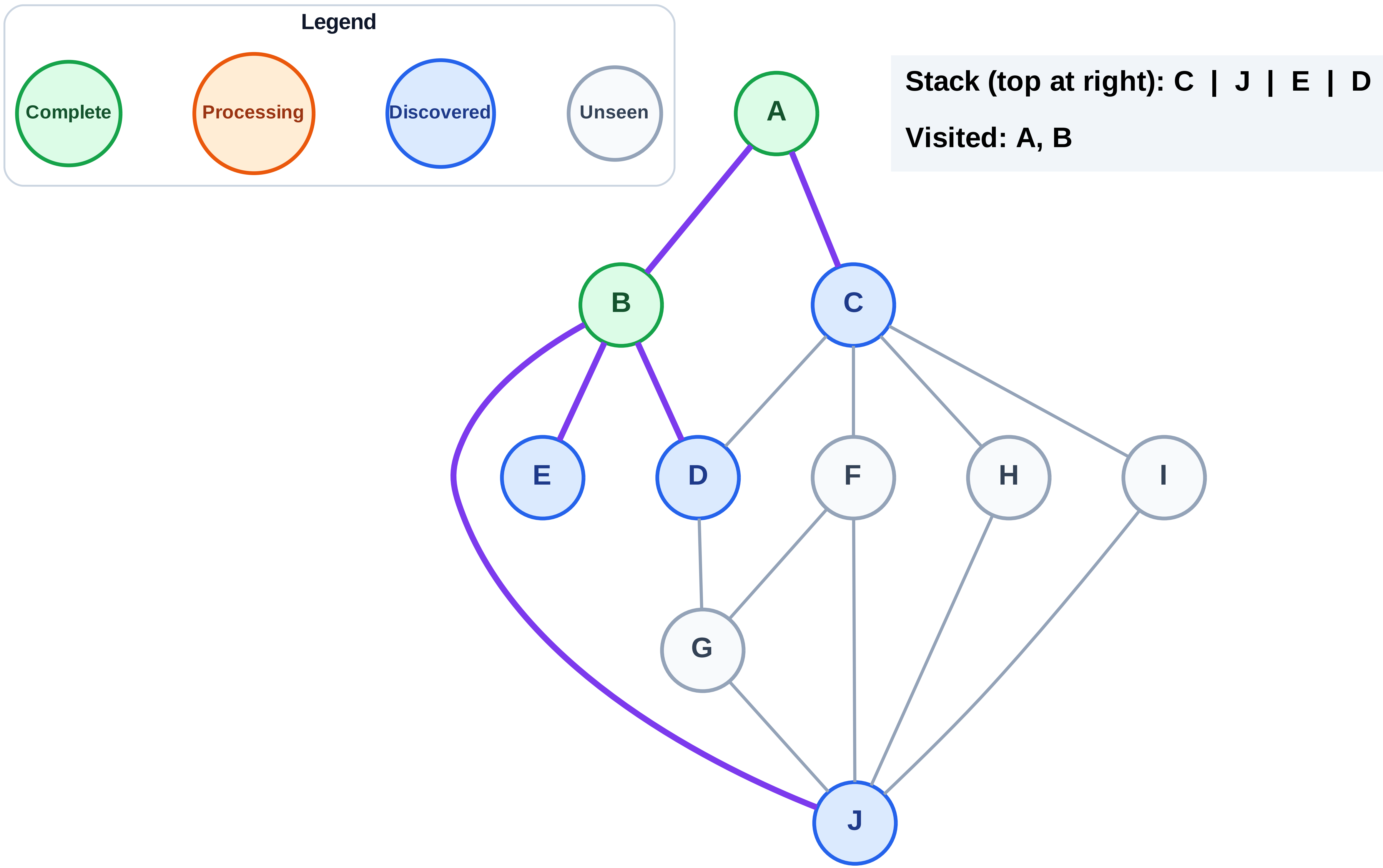
DFS Traversal

Step 4: Process node B



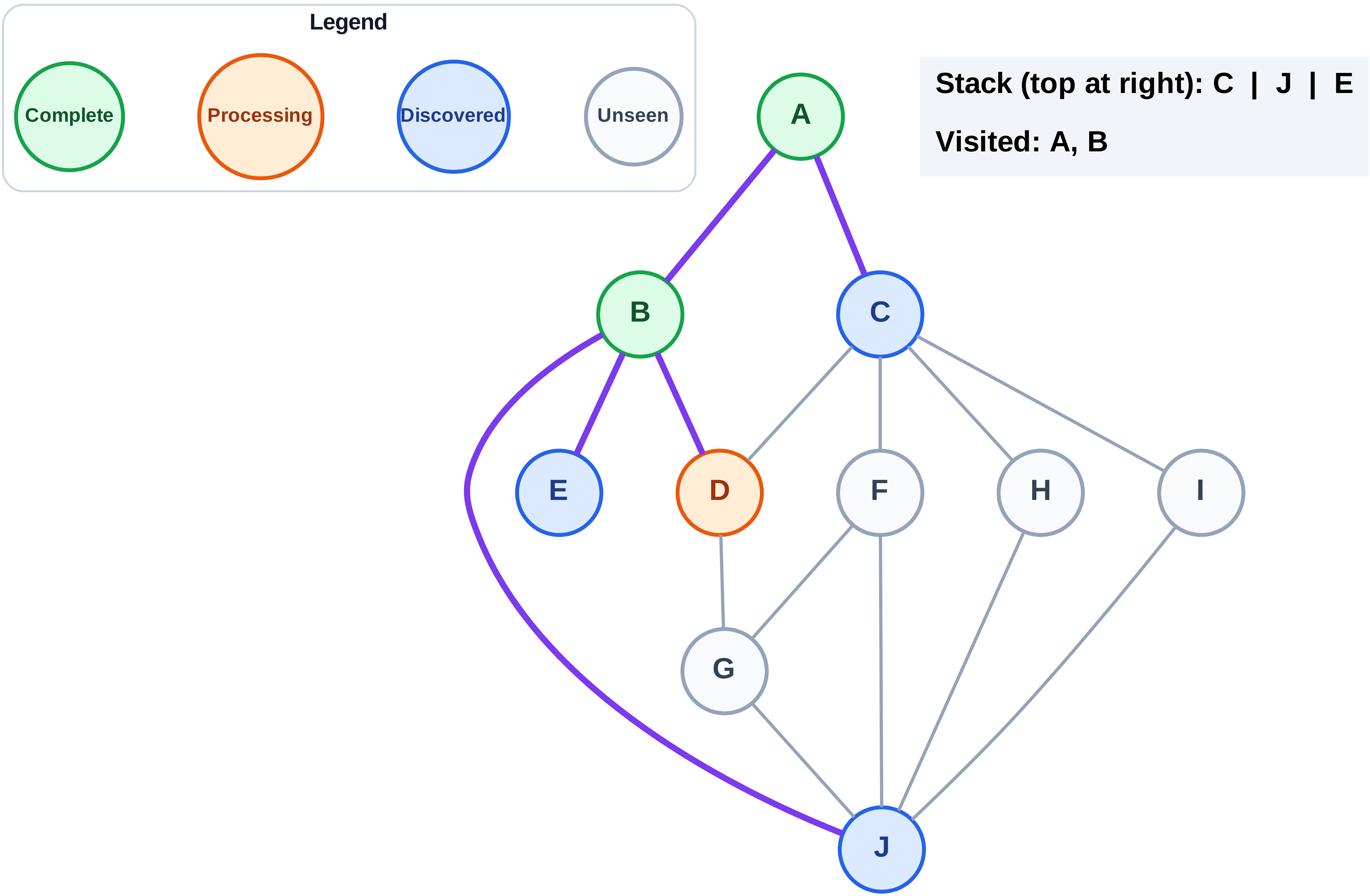
DFS Traversal

Step 5: Discover J, E, D from B



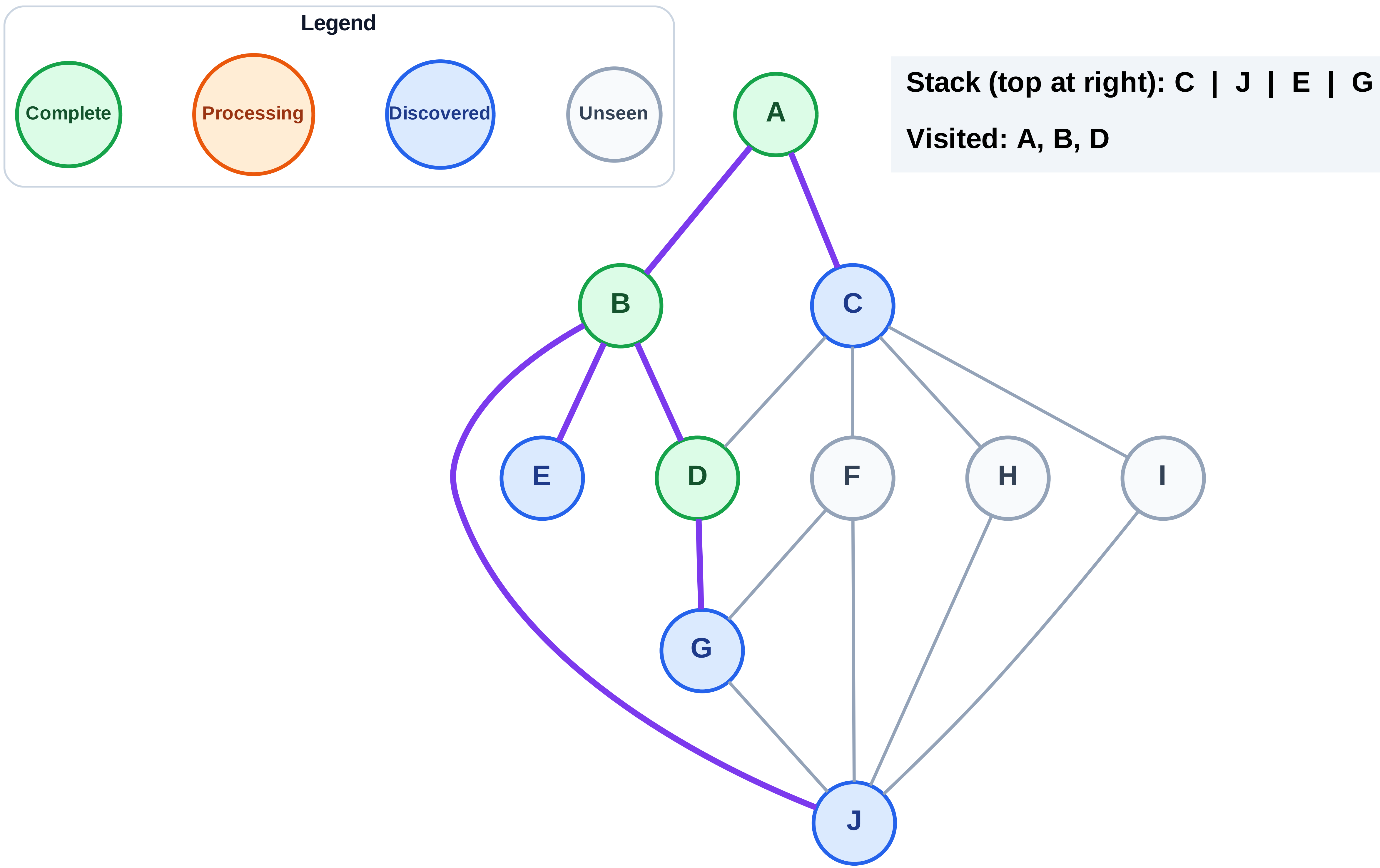
DFS Traversal

Step 6: Process node D



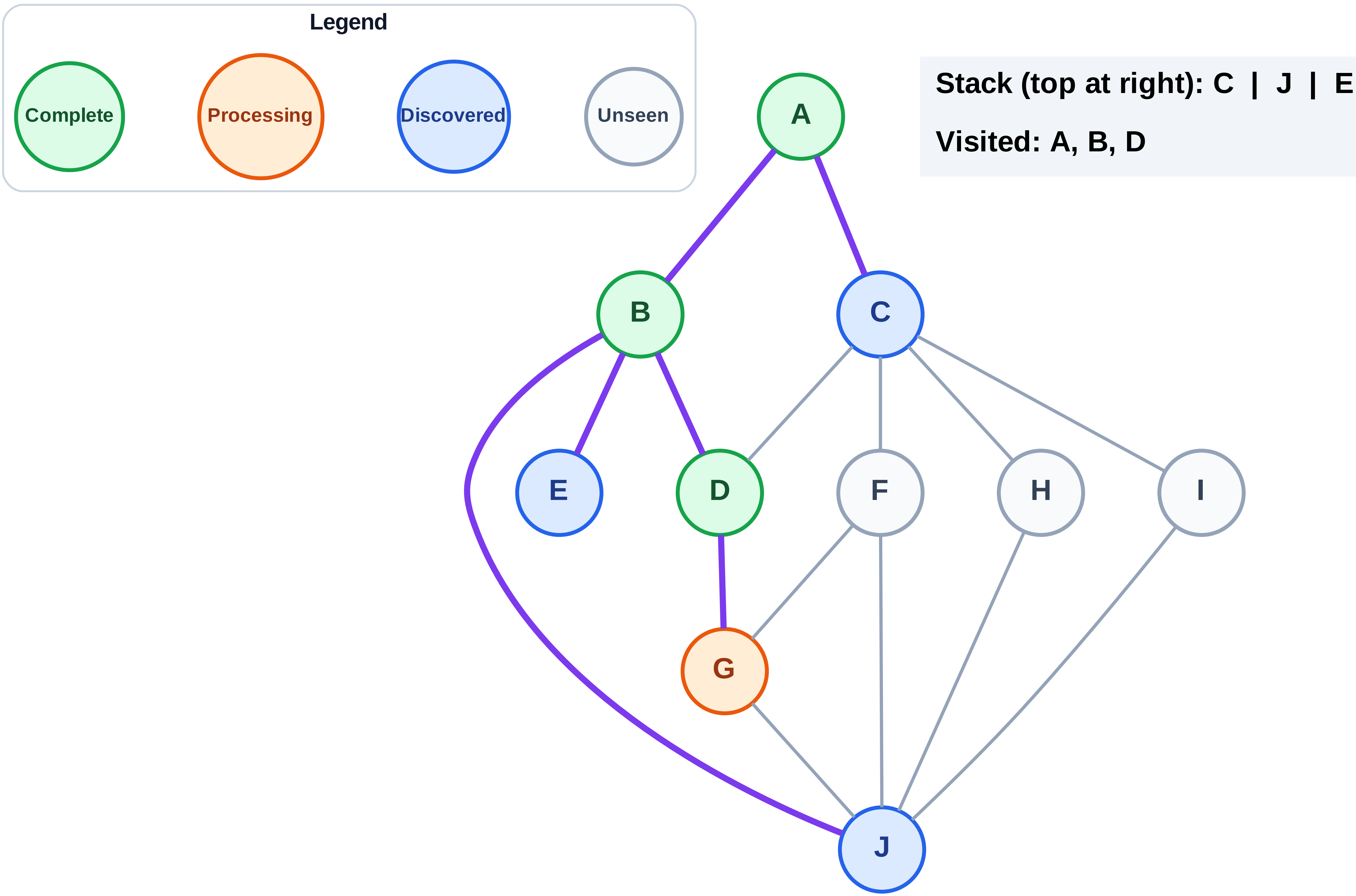
DFS Traversal

Step 7: Discover G from D



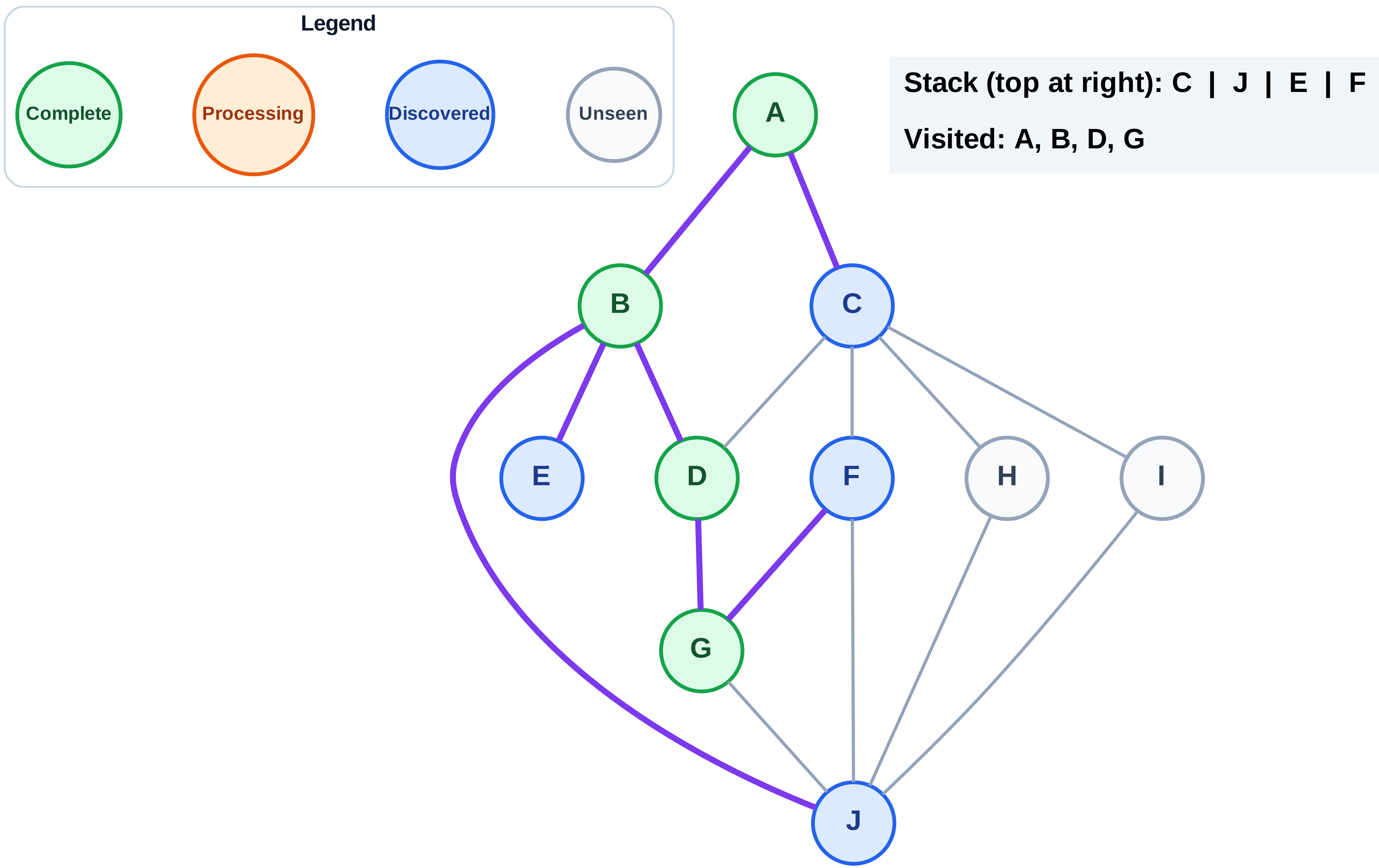
DFS Traversal

Step 8: Process node G



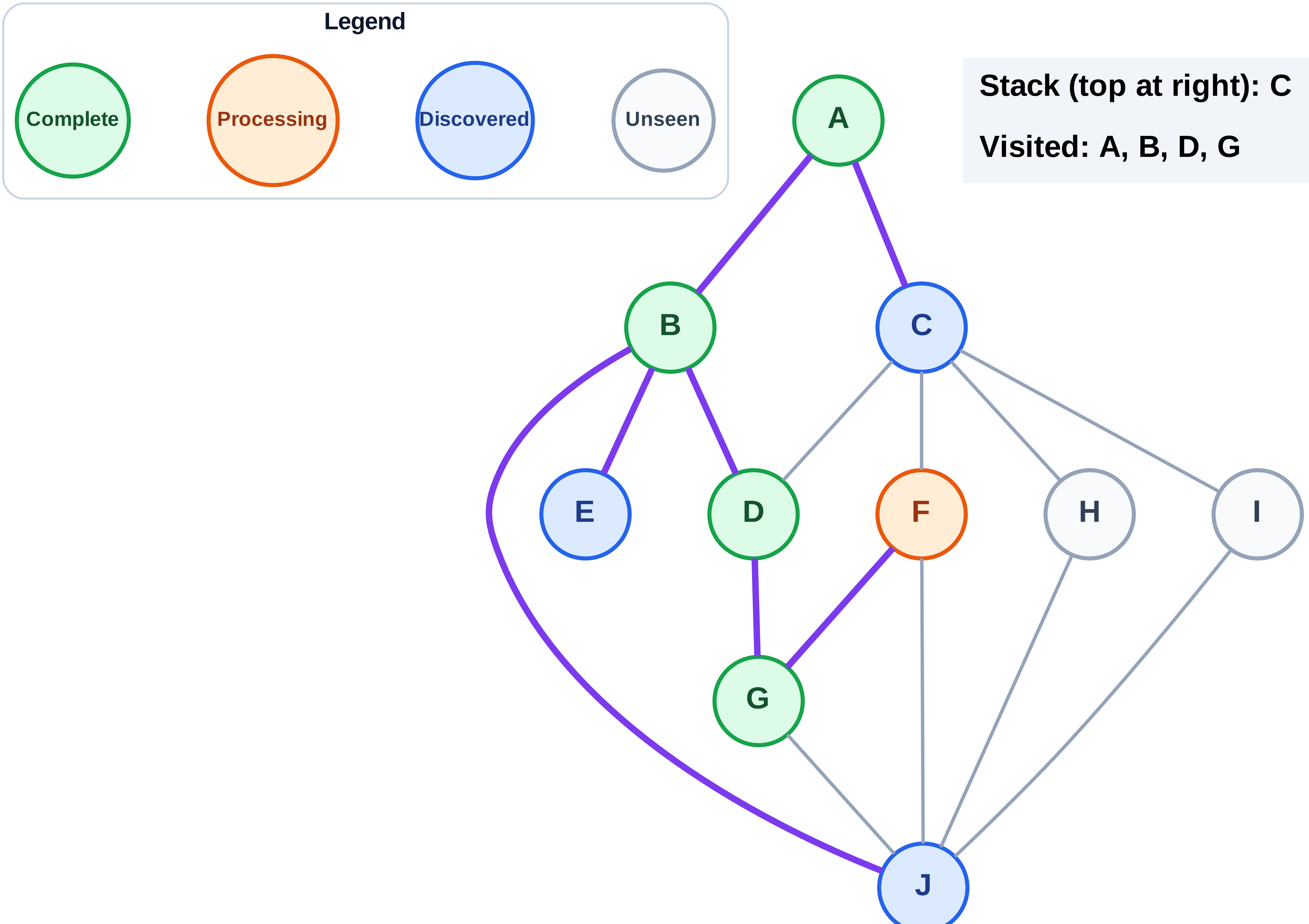
DFS Traversal

Step 9: Discover F from G



DFS Traversal

Step 10: Process node F

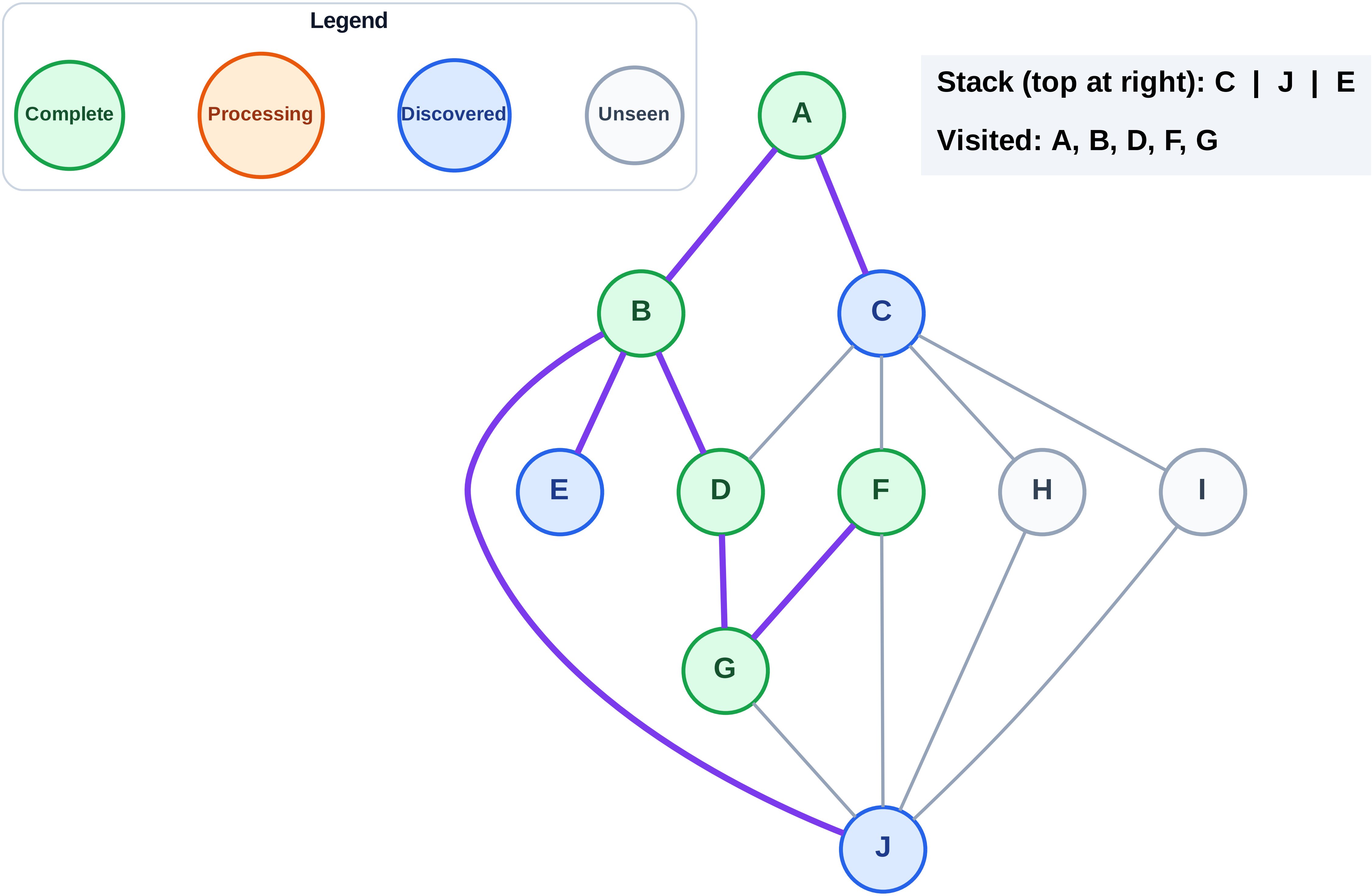


Stack (top at right): C | J | E

Visited: A, B, D, G

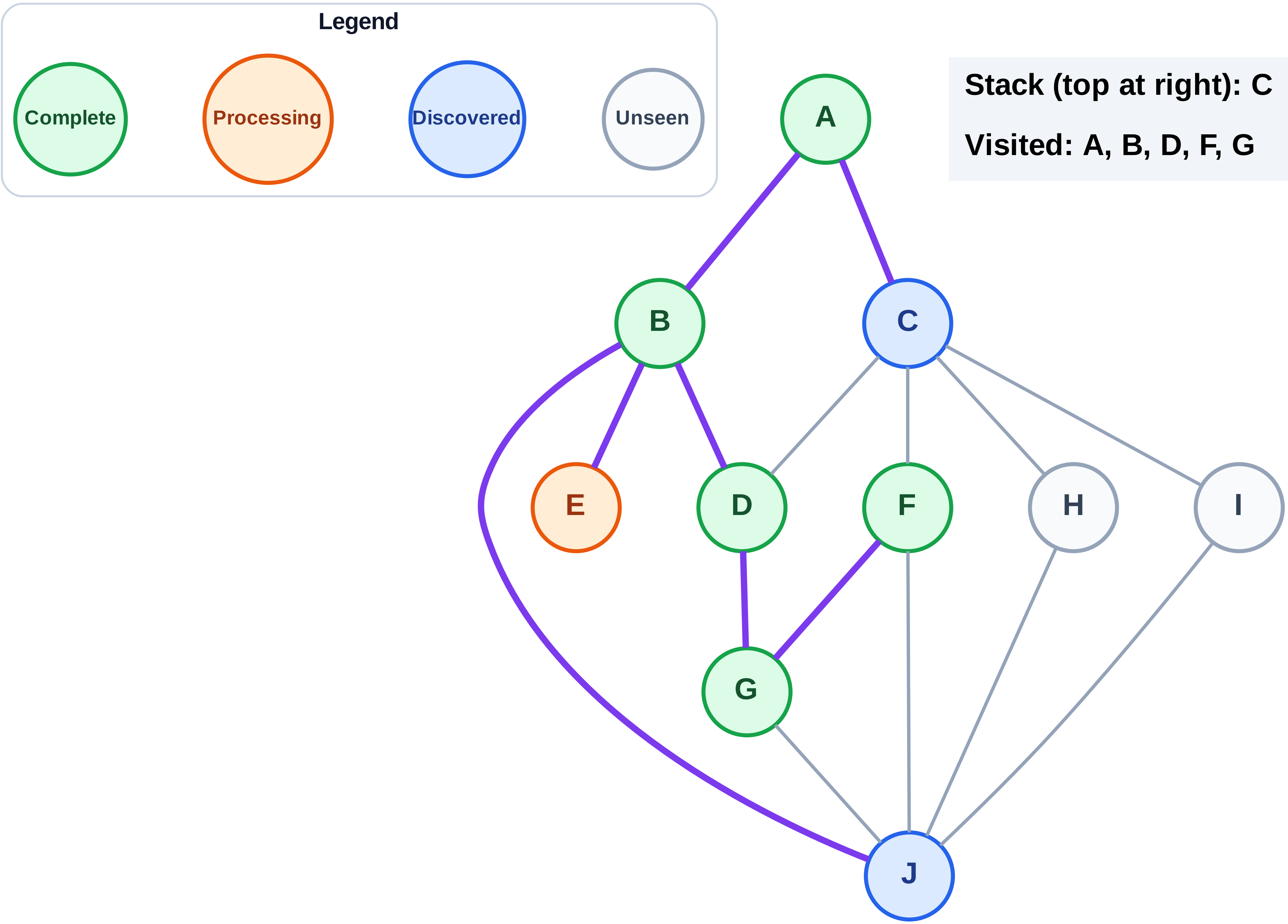
DFS Traversal

Step 11: No new neighbors from F



DFS Traversal

Step 12: Process node E

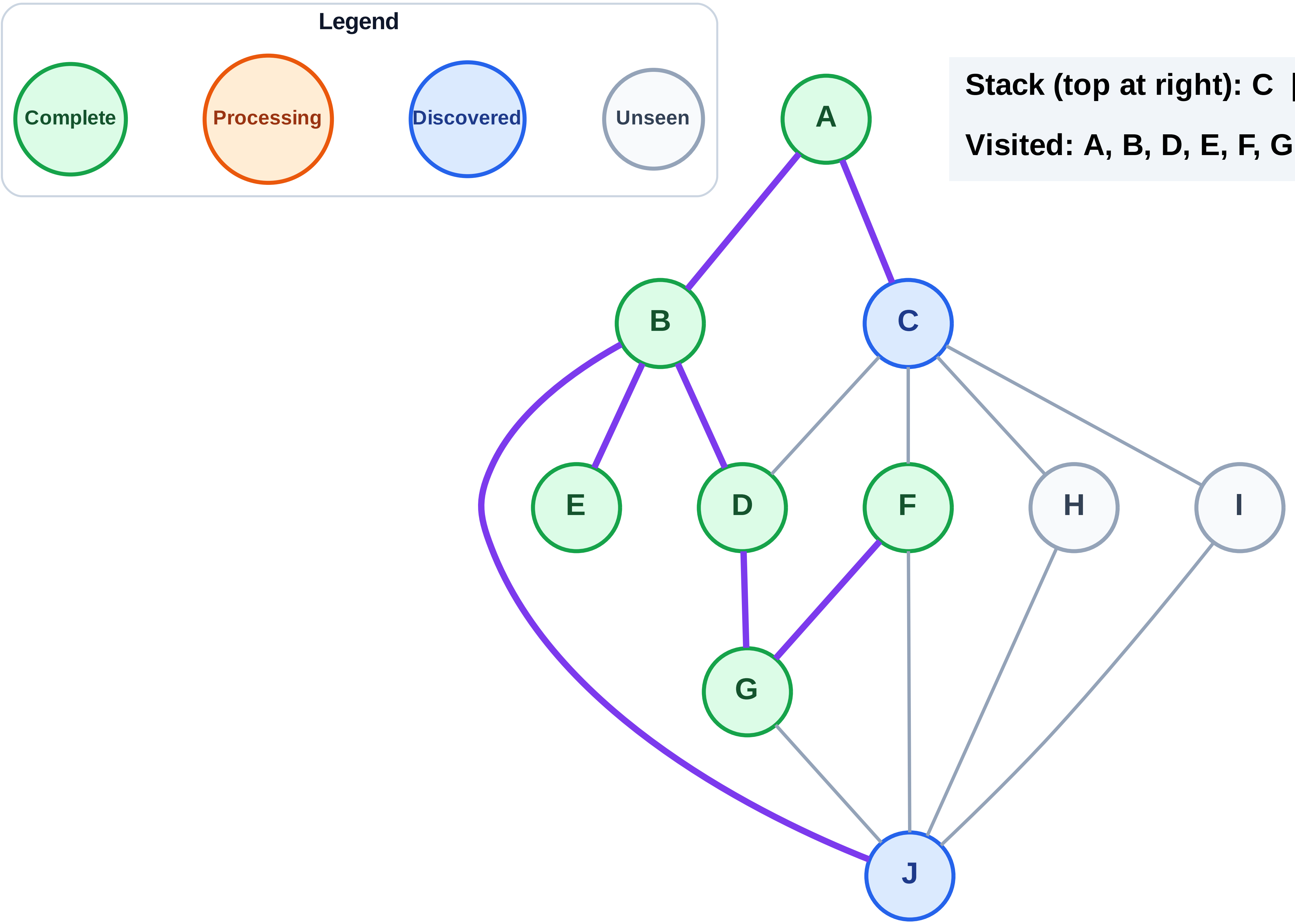


Stack (top at right): C | J

Visited: A, B, D, F, G

DFS Traversal

Step 13: No new neighbors from E

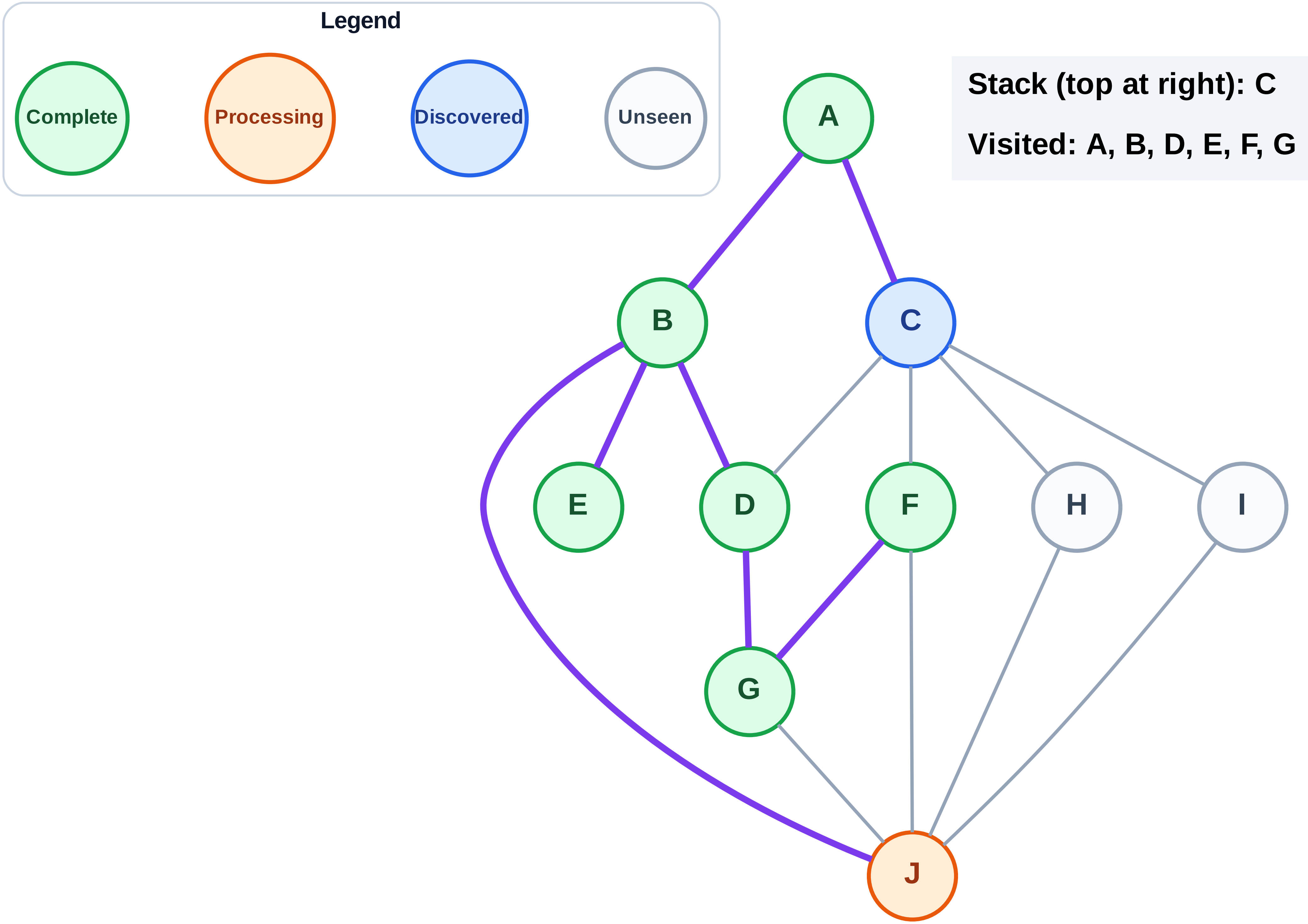


Stack (top at right): C | J

Visited: A, B, D, E, F, G

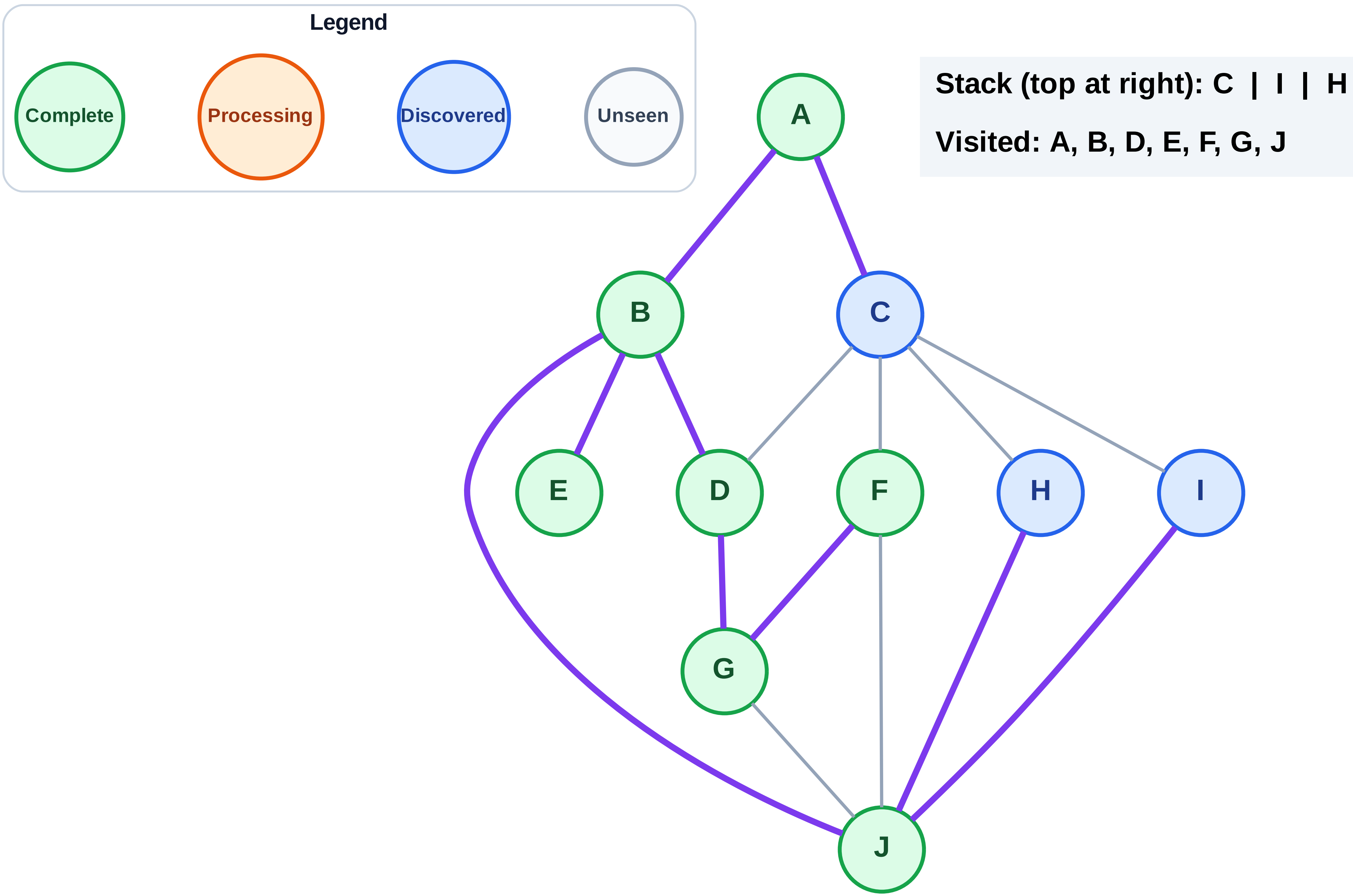
DFS Traversal

Step 14: Process node J



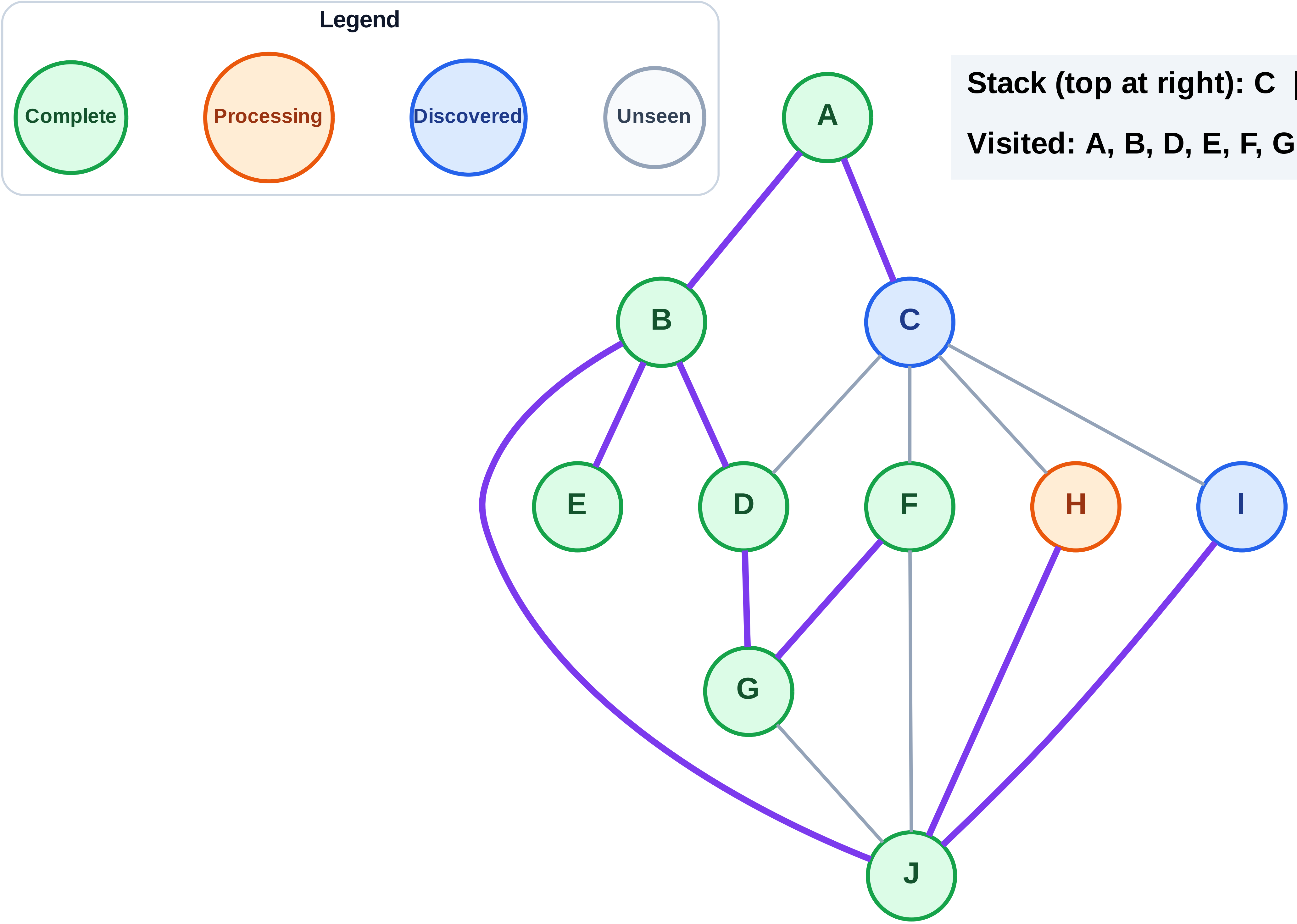
DFS Traversal

Step 15: Discover I, H from J



DFS Traversal

Step 16: Process node H

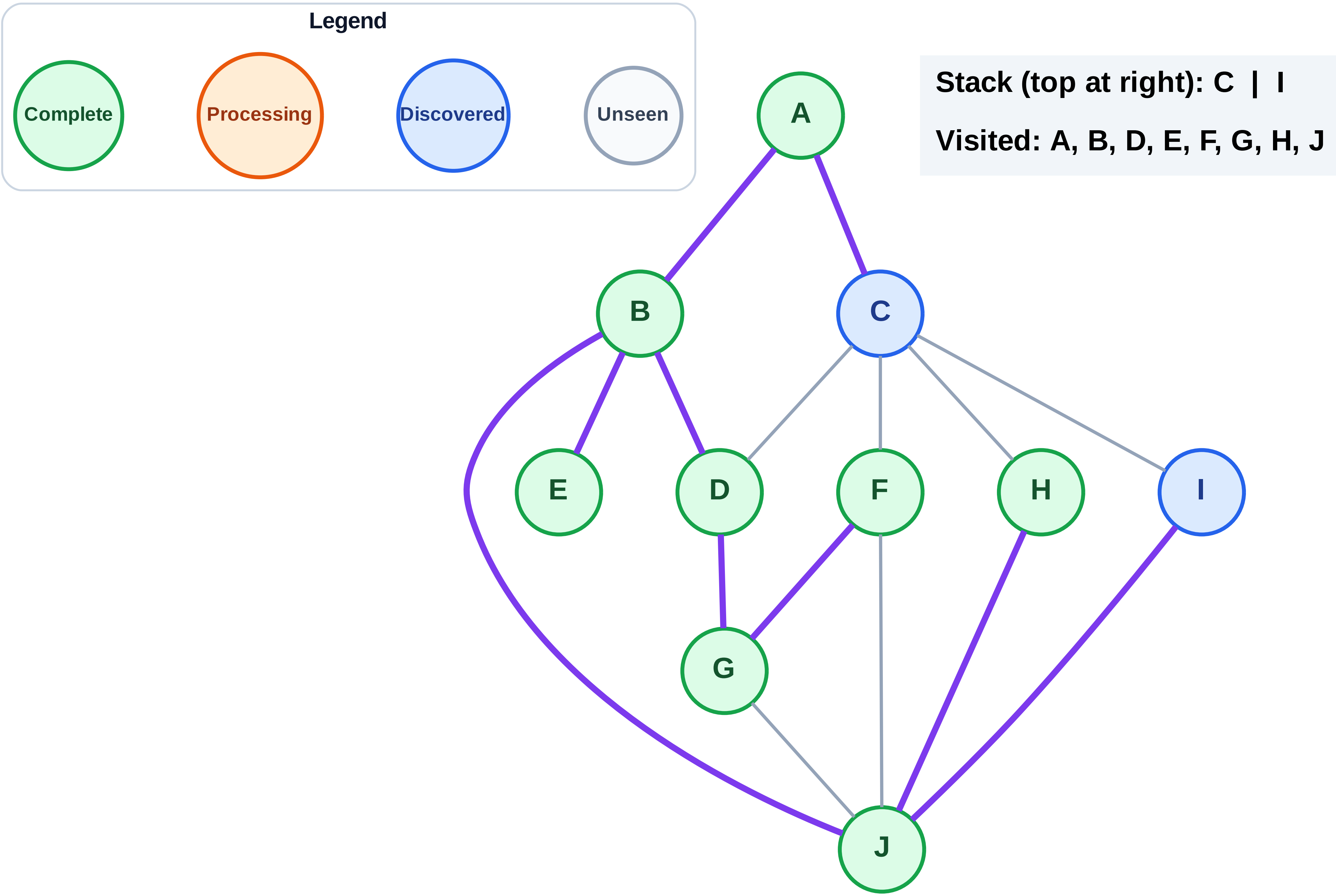


Stack (top at right): C | I

Visited: A, B, D, E, F, G, J

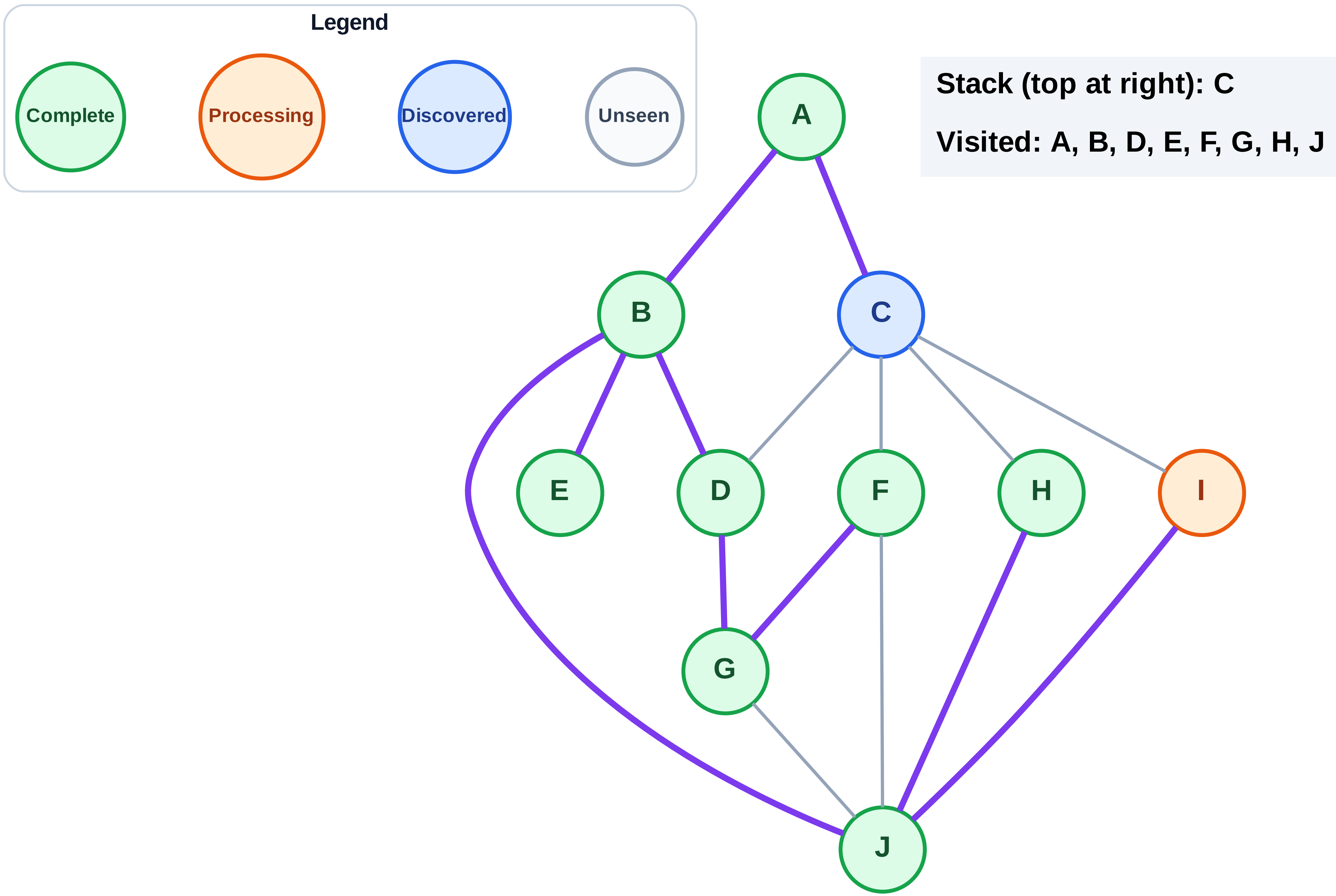
DFS Traversal

Step 17: No new neighbors from H



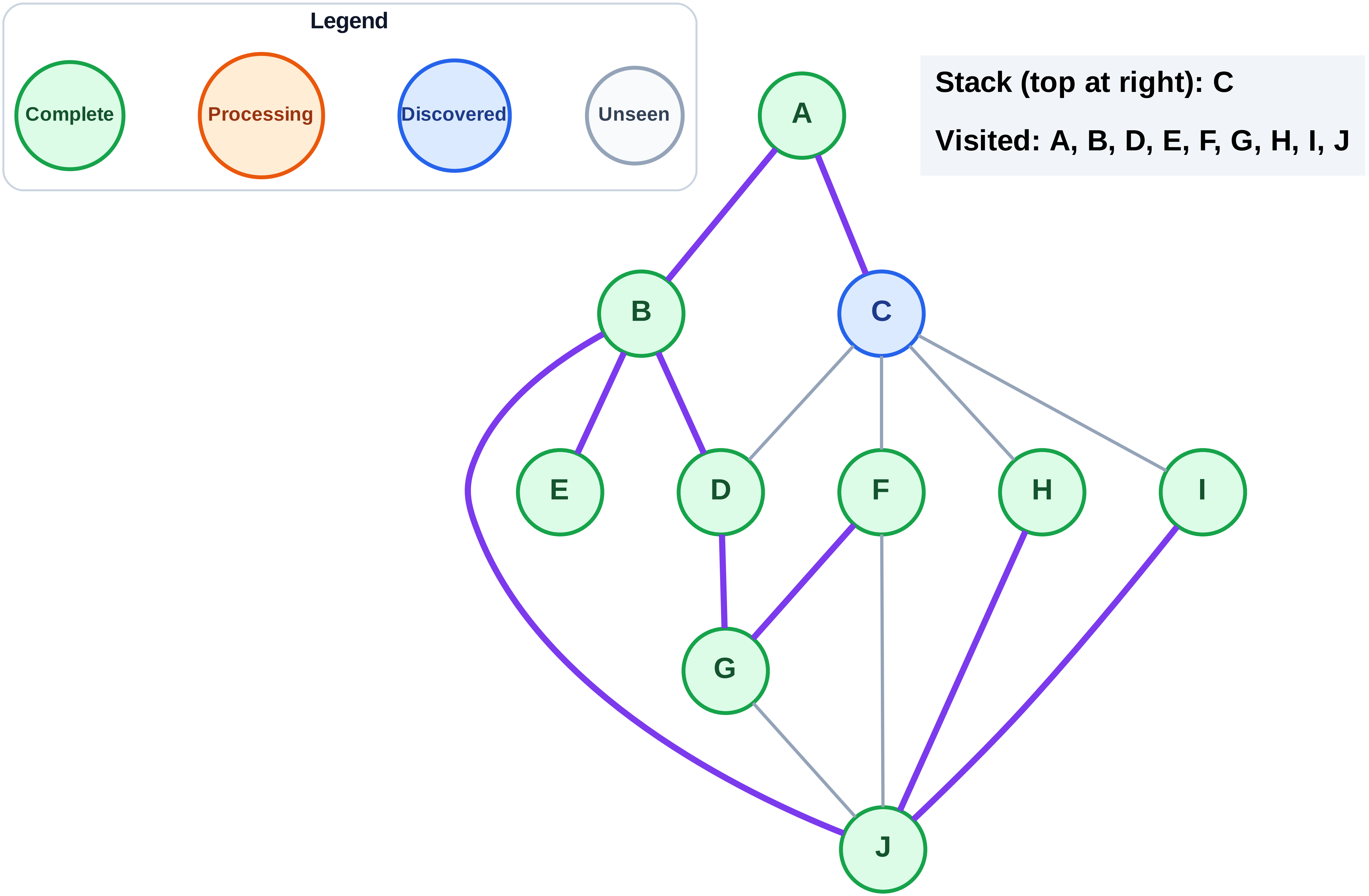
DFS Traversal

Step 18: Process node I



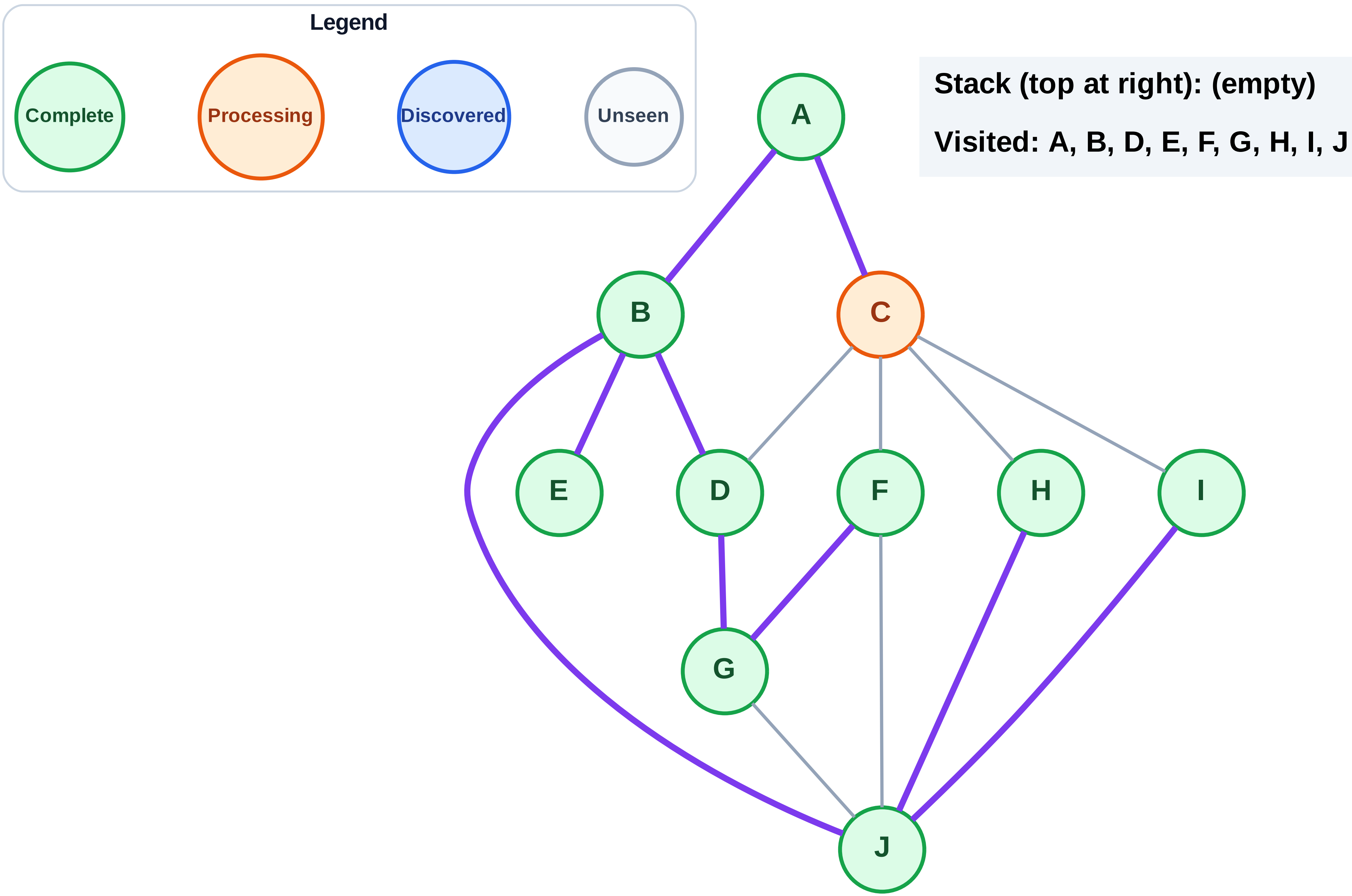
DFS Traversal

Step 19: No new neighbors from I



DFS Traversal

Step 20: Process node C



DFS Traversal

Step 21: No new neighbors from C